Schema documentation for gpcML.xsd

october 11, 2026

Table of Contents

Namespace: "" 3
Schema(s) 3
Main schema `gpcML.xsd` 3
Element(s) 3
Element `gpcML` 3
Element `gpcMLType / SummaryValues` 4
Element `SummaryValuesType / Mn` 6
Element `SummaryValuesType / Mw` 6
Element `SummaryValuesType / PDI` 6
Element `SummaryValuesType / Mz` 6
Element `SummaryValuesType / Mv` 7
Element `SummaryValuesType / n` 7
Element `SummaryValuesType / Vp` 7
Element `SummaryValuesType / Mp` 8
Element `SummaryValuesType / A` 8
Element `SummaryValuesType / AtomCount` 8
Element `SummaryValuesType / BondCount` 8
Element `gpcMLType / ContactList` 9
Element `ContactListType / Contact` 9
Element `gpcMLType / Metadata` 10
Element `MetadataType / StringRepresenation` 11
Element `StringRepListType / pSMILES` 11
Element `StringRepListType / BigSMILES` 11
Element `StringRepListType / G-BigSMILES` 11
Element `gpcMLType / InstrumentInfoList` 12
Element `InstrumentListType / RIDetectorInfo` 12
Element `RIDetectorInfoType / IDinfo` 12
Element `InstrumentListType / UVvisInfo` 13
Element `UVvisInfoType / IDinfo` 13
Element `InstrumentListType / LightScatteringInfo` 14
Element `LightScatteringInfoType / IDinfo` 14
Element `InstrumentListType / ViscometerInfo` 15
Element `ViscometerInfoType / IDinfo` 15
Element `InstrumentListType / ColumnInfo` 16
Element `ColumnInfoListType / IDinfo` 16
Element `gpcMLType / RunTimeData` 17
Element `RuntimeDataType / InstrumentConfiguration` 17
Element `InstrumentConfigurationType / ColumnList` 18
Element `ColumnListConfigurationType / GuardColumn` 18
Element `ColumnListConfigurationType / Column` 19
Element `InstrumentConfigurationType / OvenList` 20
Element `OvenListType / ColumnOven` 20
Element `OvenListType / gpcOven` 20
Element `InstrumentConfigurationType / Pump` 21
Element `InstrumentConfigurationType / Injector` 21
Element `InstrumentConfigurationType / Eluent` 21
Element `EluentType / SolventName` 22
Element `EluentType / AdditiveName` 22
Element `EluentType / RecyclingMode` 22
Element `EluentType / Ratio` 23
Element `EluentType / AdditiveConcentration` 23
Element `RuntimeDataType / RawData` 23
Element `RuntimeDataType / integrationValues` 24
Element `integrationValuesListType / Baseline` 24
Element `integrationValuesListType / Peaks` 25
Element `integrationValuesListType / Solvent` 25
Element `gpcMLType / CalibrationData` 26
Element `CalibrationDataType / TypeOfFit` 26
Element `CalibrationDataType / CoeffList` 26
Element `CoeffListType / QuinticCoeff` 27
Element `CoeffListType / QuarticCoeff` 27
Element `CoeffListType / CubicCoeff` 27
Element `CoeffListType / QuadraticCoeff` 28

Element CoeffListType / LinearCoeff ... 28
Element CoeffListType / constant ... 28
Element CalibrationDataType / MHKS_List ... 28
Element MHKSListType / K ... 29
Element MHKSListType / a ... 29
Complex Type(s) ... 29
Complex Type gpcMLType ... 29
Complex Type SummaryValuesType ... 30
Complex Type ContactListType ... 32
Complex Type ContactType ... 32
Complex Type MetadataType ... 33
Complex Type StringRepListType ... 34
Complex Type InstrumentListType ... 34
Complex Type RIDetectorInfoType ... 35
Complex Type InstrumentInfoType ... 35
Complex Type UVvisInfoType ... 36
Complex Type LightScatteringInfoType ... 36
Complex Type ViscometerInfoType ... 37
Complex Type ColumnInfoListType ... 37
Complex Type RuntimeDataType ... 37
Complex Type InstrumentConfigurationType ... 38
Complex Type ColumnListConfigurationType ... 38
Complex Type ColumnConfigurationType ... 39
Complex Type OvenListType ... 39
Complex Type OvenConfigurationType ... 39
Complex Type PumpConfigurationType ... 40
Complex Type InjectorConfigurationType ... 40
Complex Type EluentType ... 40
Complex Type RawDataType ... 41
Complex Type integrationValuesListType ... 42
Complex Type integrationValuesType ... 42
Complex Type CalibrationDataType ... 42
Complex Type CoeffListType ... 43
Complex Type MHKSListType ... 44
Complex Type Sample ... 44
Complex Type SampleType ... 44
Complex Type EluGramType ... 44
Complex Type MetadataListType ... 44
Complex Type CalibrationType ... 45
Complex Type RIDetectorRunType ... 45
Complex Type ViscometerRunType ... 45
Complex Type UVvisRunType ... 45
Complex Type LightScatteringRunType ... 45
Complex Type UVvisConfigurationType ... 45
Complex Type ColumnInfoType ... 45
Complex Type RIDetectorConfigurationType ... 46
Complex Type LightScatteringConfigurationType ... 46
Complex Type GuardColumnConfigurationType ... 46
Complex Type PolyCoeffType ... 46
Complex Type StandardUsedType ... 46
Complex Type RangeOfStandardsType ... 46
Complex Type CalibrationDateType ... 47
Simple Type(s) ... 47
Simple Type PositiveFloatType ... 47
Simple Type RatioType ... 47
Simple Type DegreePolyType ... 47
Simple Type CoeffType ... 48
Attribute(s) ... 48
Attribute ContactType / @FirstName ... 48
Attribute ContactType / @MiddleName ... 48
Attribute ContactType / @LastName ... 48
Attribute ContactType / @E-mail ... 48
Attribute ContactType / @Institution ... 49
Attribute MetadataType / @date ... 49
Attribute MetadataType / @MethodName ... 49
Attribute MetadataType / @CalibrationName ... 49
Attribute MetadataType / @Operator ... 49
Attribute MetadataType / @InstrumentName ... 49
Attribute MetadataType / @RunTime ... 50
Attribute MetadataType / @lastServiceDate ... 50
Attribute MetadataType / @SampleName ... 50
Attribute MetadataType / @ObservationalData ... 50

Attribute InstrumentInfoType / @Name 50
Attribute InstrumentInfoType / @SerialNumber 51
Attribute InstrumentInfoType / @Manufacturer 51
Attribute ColumnConfigurationType / @PoreSizeMin 51
Attribute ColumnConfigurationType / @PoreSizeMax 51
Attribute ColumnConfigurationType / @StationaryPhase 51
Attribute ColumnConfigurationType / @DateOfInstallation 51
Attribute OvenConfigurationType / @Temperature 52
Attribute PumpConfigurationType / @FlowRate 52
Attribute InjectorConfigurationType / @InjectionVolume 52
Attribute RawDataType / @Compression 52
Attribute integrationValuesType / @StartVolume 53
Attribute integrationValuesType / @EndVolume 53
Attribute gpcMLType / @Version 53
Attribute gpcMLType / @id 53
Attribute Sample / @Name 53
Attribute ColumnInfoType / @DateOfInstallation 54
Attribute ColumnInfoType / @PlateCount 54
Attribute GuardColumnConfigurationType / @ParticleSize 54
Attribute GuardColumnConfigurationType / @DateOfInstallation 54

Namespace: ""

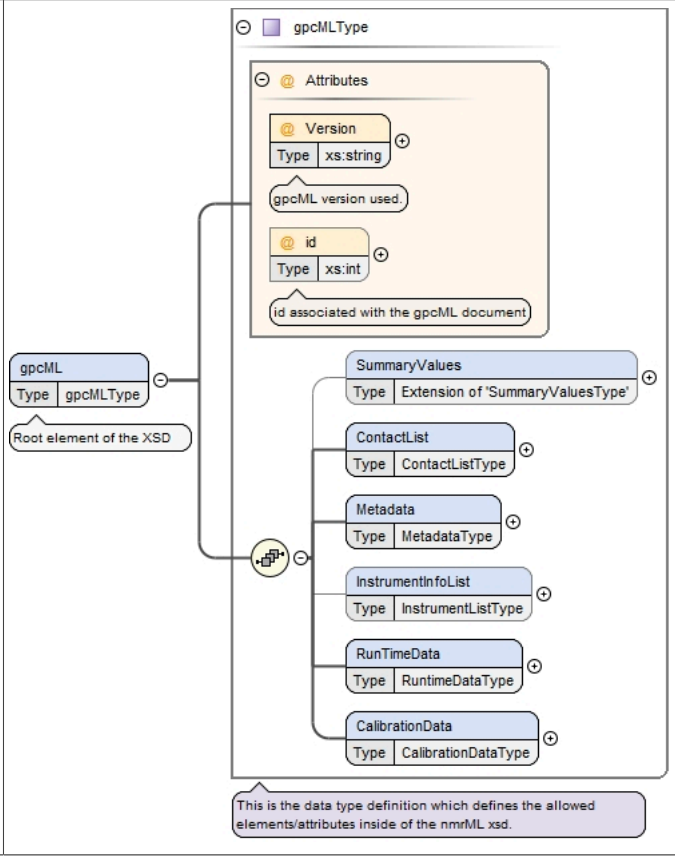
Schema(s)

Main schema gpcML.xsd

Namespace	No namespace
Properties	attribute form default: unqualified element form default: unqualified

Element(s)

Element gpcML

Namespace	No namespace
Annotations	Root element of the XSD
Diagram	

Type	gpcMLType
Properties	content: complex
Model	SummaryValues{0,1} , ContactList , Metadata , InstrumentInfoList{0,1} , RunTimeData , CalibrationData
Children	CalibrationData, ContactList, InstrumentInfoList, Metadata, RunTimeData, SummaryValues
Instance	<gpcML id="" Version=""> <SummaryValues>{0,1}</SummaryValues> <ContactList>{1,1}</ContactList> <Metadata CalibrationName="" date="" InstrumentName="" lastServiceDate="" MethodName="" ObservationalData="" Oper Metadata> <InstrumentInfoList>{0,1}</InstrumentInfoList> <RunTimeData>{1,1}</RunTimeData> <CalibrationData>{1,1}</CalibrationData> </gpcML>

Attributes

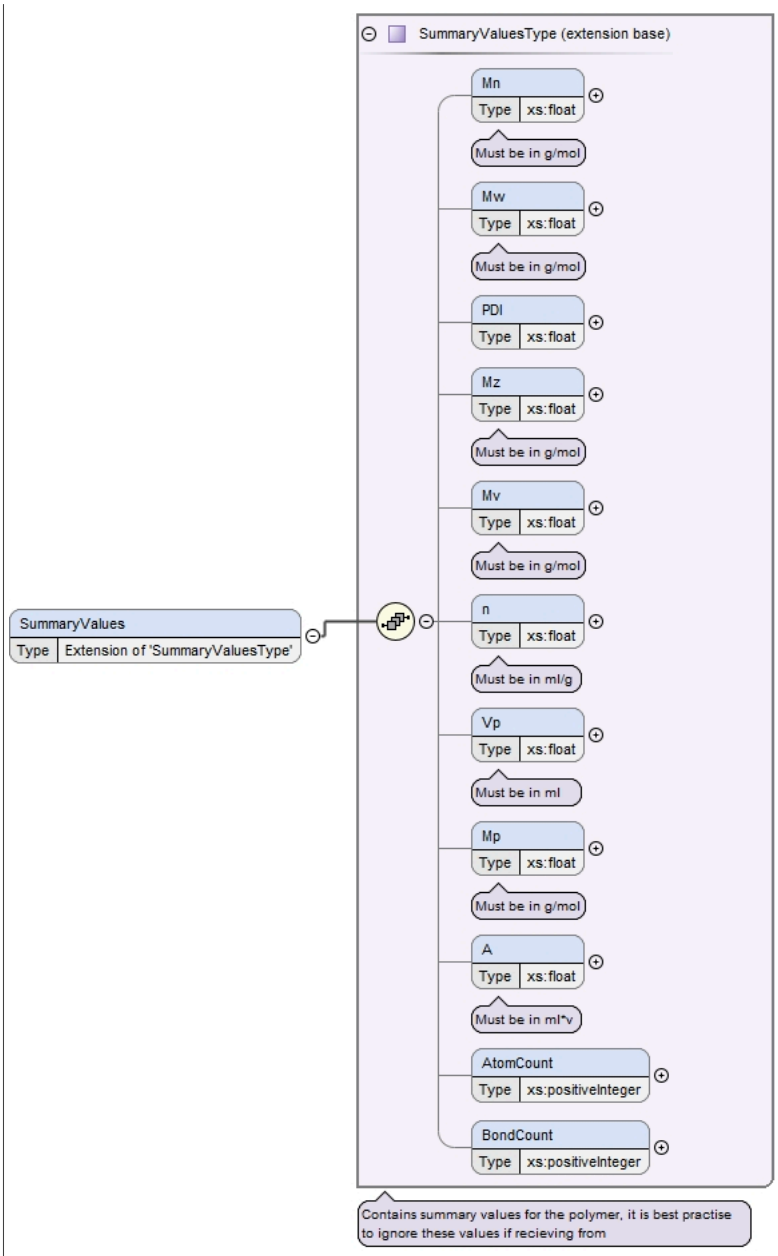
QName	Type	Use	
Version	xs:string	required	
	gpcML version used.		
id	xs:int	optional	
	id associated with the gpcML document		

Source

```
<xs:element name="gpcML" type="gpcMLType">
  <xs:annotation>
    <xs:documentation>Root element of the XSD</xs:documentation>
  </xs:annotation>
</xs:element>
```

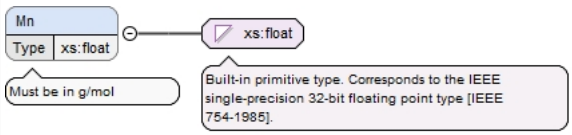
Element gpcMLType / SummaryValues

Namespace	No namespace

Diagram	
Type	extension of SummaryValuesType
Type hierarchy	• SummaryValuesType
Properties	content: complex minOccurs: 0
Model	Mn{0,1} , Mw{0,1} , PDI{0,1} , Mz{0,1} , Mv{0,1} , n{0,1} , Vp{0,1} , Mp{0,1} , A{0,1} , AtomCount{0,1} , BondCount{0,1}
Children	A, AtomCount, BondCount, Mn, Mp, Mv, Mw, Mz, PDI, Vp, n
Instance	<SummaryValues> <Mn>{0,1}</Mn> <Mw>{0,1}</Mw> <PDI>{0,1}</PDI> <Mz>{0,1}</Mz> <Mv>{0,1}</Mv> <n>{0,1}</n> <Vp>{0,1}</Vp> <Mp>{0,1}</Mp> <A>{0,1}</A> <AtomCount>{0,1}</AtomCount> <BondCount>{0,1}</BondCount> </SummaryValues>
Source	<xs:element name="SummaryValues" minOccurs="0">

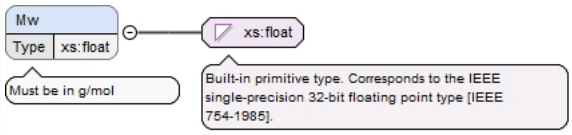
```
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="SummaryValuesType"/>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
```

Element SummaryValuesType / Mn

Namespace	No namespace
Annotations	Must be in g/mol
Diagram	
Type	xs:float
Properties	content: simple minOccurs: 0

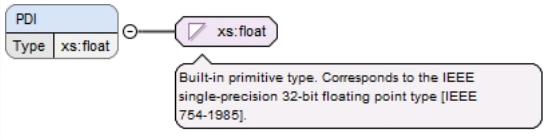
Source	`<xs:element name="Mn" minOccurs="0" type="xs:float">`
	`  <xs:annotation>`
	`  <xs:documentation>Must be in g/mol</xs:documentation>`
	`  </xs:annotation>`
	`</xs:element>`

Element SummaryValuesType / Mw

Namespace	No namespace
Annotations	Must be in g/mol
Diagram	
Type	xs:float
Properties	content: simple minOccurs: 0

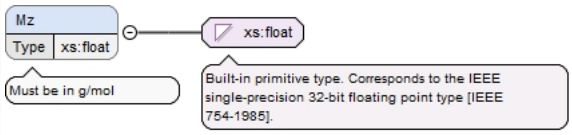
Source	`<xs:element name="Mw" minOccurs="0" type="xs:float">`
	`  <xs:annotation>`
	`  <xs:documentation>Must be in g/mol</xs:documentation>`
	`  </xs:annotation>`
	`</xs:element>`

Element SummaryValuesType / PDI

Namespace	No namespace
Diagram	
Type	xs:float
Properties	content: simple minOccurs: 0
Source	`<xs:element name="PDI" minOccurs="0" type="xs:float"/>`

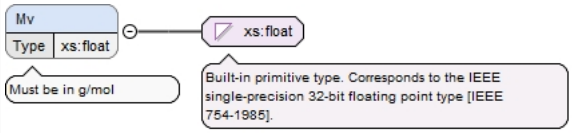
Element SummaryValuesType / Mz

Namespace	No namespace
Annotations	Must be in g/mol

Diagram	
Type	xs:float
Properties	content: simple minOccurs: 0

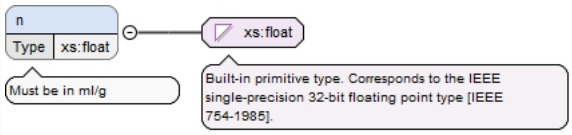
Source	`<xs:element name="Mz" minOccurs="0" type="xs:float">`
	`  <xs:annotation>`
	`  <xs:documentation>Must be in g/mol</xs:documentation>`
	`  </xs:annotation>`
	`</xs:element>`

Element SummaryValuesType / Mv

Namespace	No namespace
Annotations	Must be in g/mol
Diagram	
Type	xs:float
Properties	content: simple minOccurs: 0

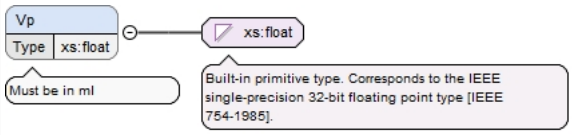
Source	`<xs:element name="Mv" minOccurs="0" type="xs:float">`
	`  <xs:annotation>`
	`  <xs:documentation>Must be in g/mol</xs:documentation>`
	`  </xs:annotation>`
	`</xs:element>`

Element SummaryValuesType / n

Namespace	No namespace
Annotations	Must be in ml/g
Diagram	
Type	xs:float
Properties	content: simple minOccurs: 0

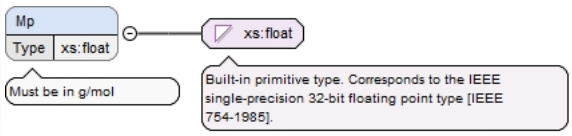
Source	`<xs:element name="n" type="xs:float" minOccurs="0">`
	`  <xs:annotation>`
	`  <xs:documentation>Must be in ml/g</xs:documentation>`
	`  </xs:annotation>`
	`</xs:element>`

Element SummaryValuesType / Vp

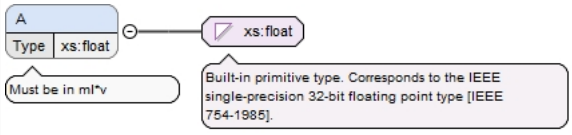
Namespace	No namespace
Annotations	Must be in ml
Diagram	
Type	xs:float

Properties	content: simple minOccurs: 0
Source	<xs:element name="Vp" minOccurs="0" type="xs:float"> <xs:annotation> <xs:documentation>Must be in ml</xs:documentation> </xs:annotation> </xs:element>

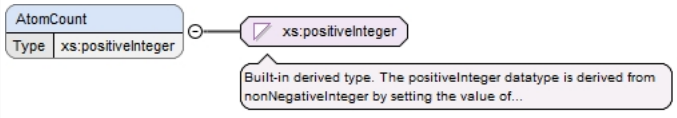
Element SummaryValuesType / Mp

Namespace	No namespace
Annotations	Must be in g/mol
Diagram	
Type	xs:float
Properties	content: simple minOccurs: 0
Source	<xs:element name="Mp" minOccurs="0" type="xs:float"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element>

Element SummaryValuesType / A

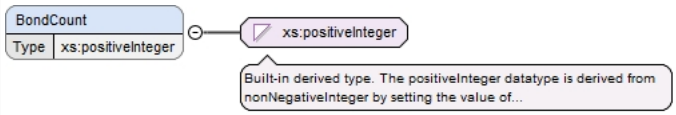
Namespace	No namespace
Annotations	Must be in ml*v
Diagram	
Type	xs:float
Properties	content: simple minOccurs: 0
Source	<xs:element name="A" minOccurs="0" type="xs:float"> <xs:annotation> <xs:documentation>Must be in ml*v</xs:documentation> </xs:annotation> </xs:element>

Element SummaryValuesType / AtomCount

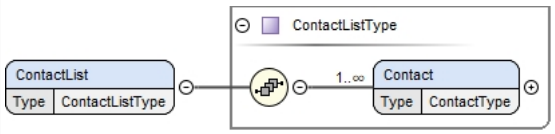
Namespace	No namespace
Diagram	
Type	xs:positiveInteger
Properties	content: simple minOccurs: 0
Source	<xs:element minOccurs="0" name="AtomCount" type="xs:positiveInteger"/>

Element SummaryValuesType / BondCount

Namespace	No namespace

Diagram	
Type	xs:positiveInteger
Properties	content: simple minOccurs: 0
Source	`<xs:element minOccurs="0" name="BondCount" type="xs:positiveInteger"/>`

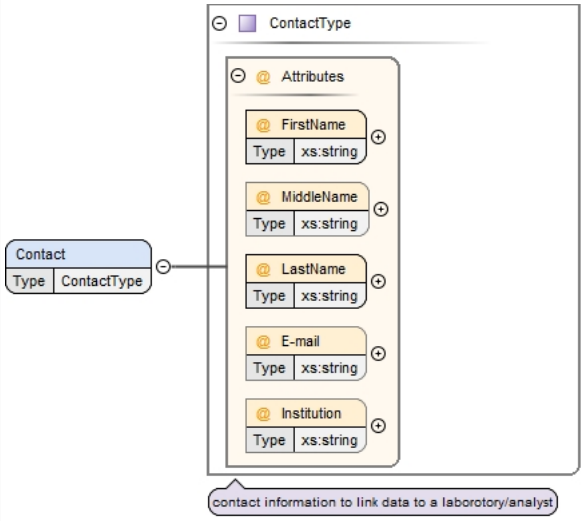
Element gpcMLType / ContactList

Namespace	No namespace
Diagram	
Type	ContactListType
Properties	content: complex
Model	Contact+
Children	Contact

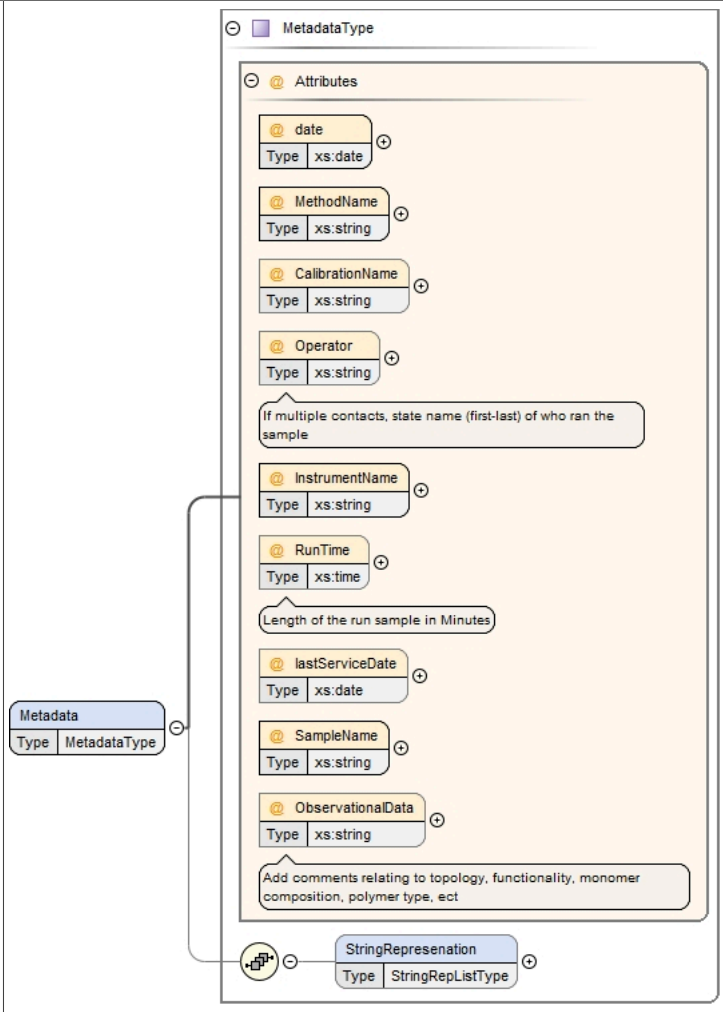
Instance	`<ContactList>`
	`  <Contact E-mail="" FirstName="" Institution="" LastName="" MiddleName="">{1,unbounded}</Contact>`
	`</ContactList>`

Source	`<xs:element name="ContactList" type="ContactListType"/>`

Element ContactListType / Contact

Namespace	No namespace
Diagram	
Type	ContactType
Properties	content: complex maxOccurs: unbounded
Attributes	**QName** / **Type** / **Use** **E-mail** xs:string optional **FirstName** xs:string required **Institution** xs:string optional **LastName** xs:string required **MiddleName** xs:string optional
Source	`<xs:element name="Contact" type="ContactType" maxOccurs="unbounded"/>`

Element gpcMLType / Metadata

Namespace	No namespace
Diagram	
Type	MetadataType
Properties	content: complex
Model	StringRepresenation{0,1}
Children	StringRepresenation

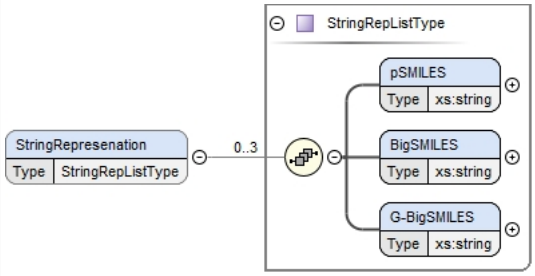
Instance	`<Metadata CalibrationName="" date="" InstrumentName="" lastServiceDate="" MethodName="" ObservationalData="" Operat`
	`  <StringRepresenation>{0,1}</StringRepresenation>`
	`</Metadata>`

Attributes

QName	Type	Use	
CalibrationName	xs:string	optional	
InstrumentName	xs:string	required	
MethodName	xs:string	required	
ObservationalData	xs:string	optional	
	Add comments relating to topology, functionality, monomer composition, polymer type, ect		
Operator	xs:string	optional	
	If multiple contacts, state name (first-last) of who ran the sample		
RunTime	xs:time	optional	
	Length of the run sample in Minutes		
SampleName	xs:string	required	
date	xs:date	required	
lastServiceDate	xs:date	optional	

Source	`<xs:element name="Metadata" type="MetadataType"/>`

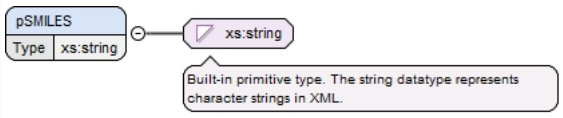
Element `MetadataType / StringRepresenation`

Namespace	No namespace
Diagram	
Type	StringRepListType
Properties	content: complex minOccurs: 0
Model	pSMILES , BigSMILES , G-BigSMILES
Children	BigSMILES, G-BigSMILES, pSMILES

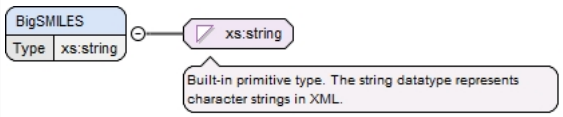
Instance	`<StringRepresenation>`
	`  <pSMILES>{1,1}</pSMILES>`
	`  <BigSMILES>{1,1}</BigSMILES>`
	`  <G-BigSMILES>{1,1}</G-BigSMILES>`
	`</StringRepresenation>`

Source	`<xs:element minOccurs="0" name="StringRepresenation" type="StringRepListType"/>`

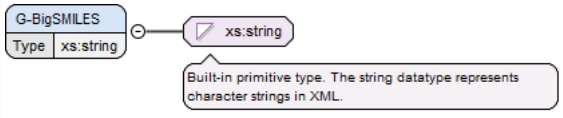
Element `StringRepListType / pSMILES`

Namespace	No namespace
Diagram	
Type	xs:string
Properties	content: simple
Source	`<xs:element name="pSMILES" type="xs:string"/>`

Element `StringRepListType / BigSMILES`

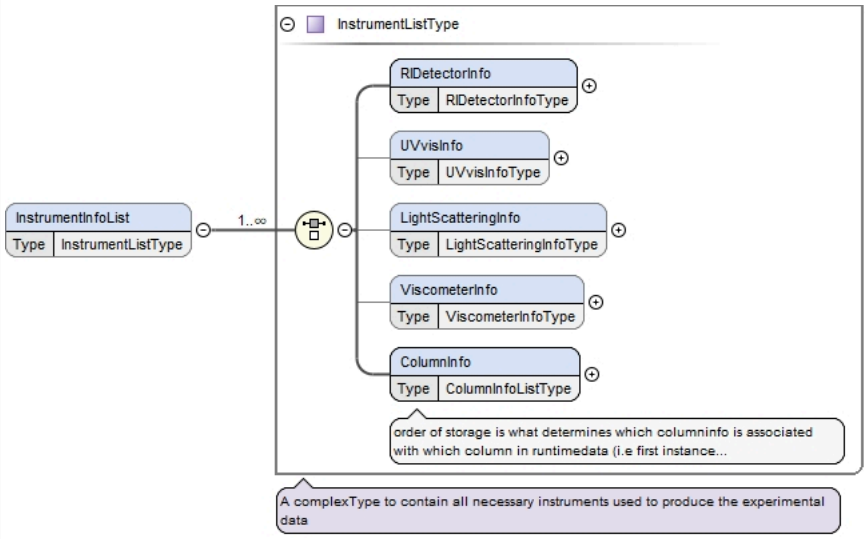
Namespace	No namespace
Diagram	
Type	xs:string
Properties	content: simple
Source	`<xs:element name="BigSMILES" type="xs:string"/>`

Element `StringRepListType / G-BigSMILES`

Namespace	No namespace
Diagram	
Type	xs:string

Properties	content: simple
Source	`<xs:element name="G-BigSMILES" type="xs:string"/>`

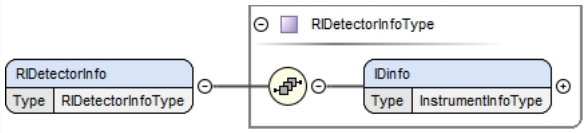
Element gpcMLType / InstrumentInfoList

Namespace	No namespace
Diagram	
Type	InstrumentListType
Properties	content: complex minOccurs: 0
Model	RIDetectorInfo | UVvisInfo{0,1} | LightScatteringInfo{0,1} | ViscometerInfo{0,1} | ColumnInfo
Children	ColumnInfo, LightScatteringInfo, RIDetectorInfo, UVvisInfo, ViscometerInfo

Instance	`<InstrumentInfoList>`
	`  <RIDetectorInfo>{1,1}</RIDetectorInfo>`
	`  <UVvisInfo>{0,1}</UVvisInfo>`
	`  <LightScatteringInfo>{0,1}</LightScatteringInfo>`
	`  <ViscometerInfo>{0,1}</ViscometerInfo>`
	`  <ColumnInfo>{1,1}</ColumnInfo>`
	`</InstrumentInfoList>`

Source	`<xs:element name="InstrumentInfoList" type="InstrumentListType" minOccurs="0"/>`

Element InstrumentListType / RIDetectorInfo

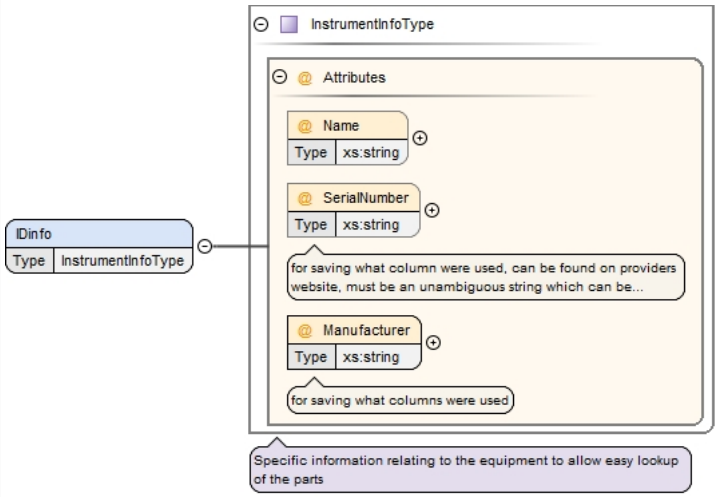
Namespace	No namespace
Diagram	
Type	RIDetectorInfoType
Properties	content: complex
Model	IDinfo
Children	IDinfo

Instance	`<RIDetectorInfo>`
	`  <IDinfo Manufacturer="" Name="" SerialNumber="">{1,1}</IDinfo>`
	`</RIDetectorInfo>`

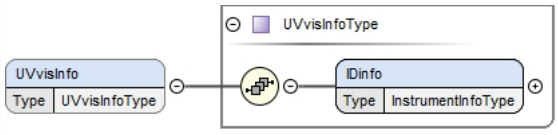
Source	`<xs:element name="RIDetectorInfo" type="RIDetectorInfoType"/>`

Element RIDetectorInfoType / IDinfo

Namespace	No namespace

Diagram	
Type	InstrumentInfoType
Properties	content: complex
Attributes	**QName** / **Type** / **Use** **Manufacturer** — xs:string — required for saving what columns were used **Name** — xs:string — optional **SerialNumber** — xs:string — optional for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.
Source	`<xs:element name="IDinfo" type="InstrumentInfoType"/>`

Element `InstrumentListType / UVvisInfo`

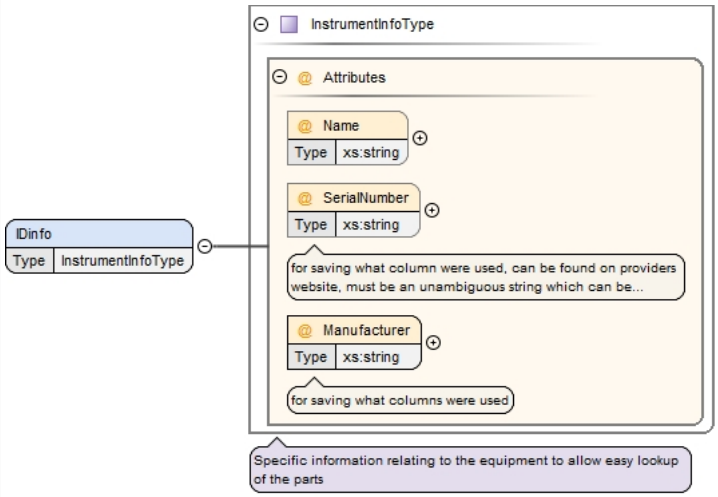
Namespace	No namespace
Diagram	
Type	UVvisInfoType
Properties	content: complex minOccurs: 0
Model	IDinfo
Children	IDinfo

Instance	`<UVvisInfo>`
	`  <IDinfo Manufacturer="" Name="" SerialNumber="">{1,1}</IDinfo>`
	`</UVvisInfo>`

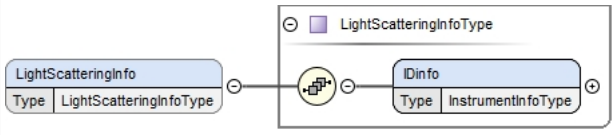
Source	`<xs:element name="UVvisInfo" type="UVvisInfoType" minOccurs="0"/>`

Element `UVvisInfoType / IDinfo`

Namespace	No namespace

Diagram	
Type	InstrumentInfoType
Properties	content: complex
Attributes	**QName** / **Type** / **Use** **Manufacturer** — xs:string — required for saving what columns were used **Name** — xs:string — optional **SerialNumber** — xs:string — optional for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.
Source	`<xs:element name="IDinfo" type="InstrumentInfoType"/>`

Element `InstrumentListType / LightScatteringInfo`

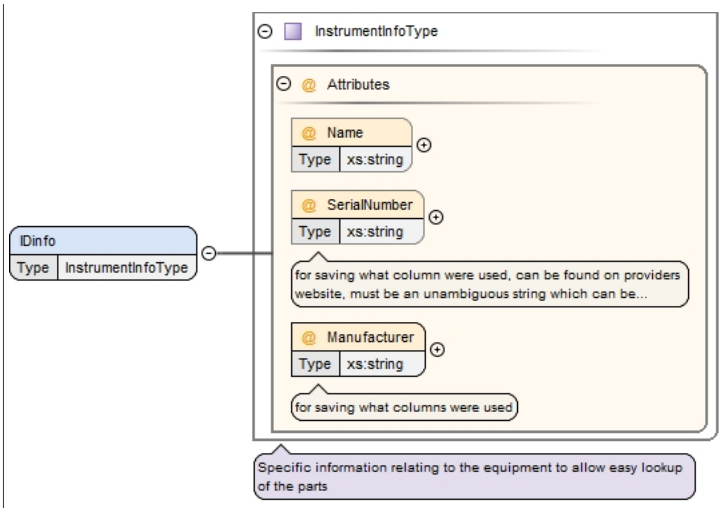
Namespace	No namespace
Diagram	
Type	LightScatteringInfoType
Properties	content: complex minOccurs: 0
Model	IDinfo
Children	IDinfo

Instance	`<LightScatteringInfo>`
	`  <IDinfo Manufacturer="" Name="" SerialNumber="">{1,1}</IDinfo>`
	`</LightScatteringInfo>`

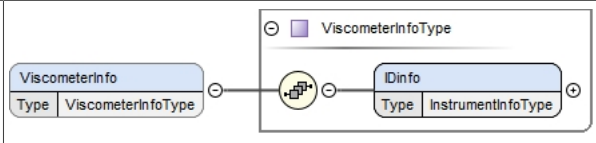
Source	`<xs:element name="LightScatteringInfo" type="LightScatteringInfoType" minOccurs="0"/>`

Element `LightScatteringInfoType / IDinfo`

Namespace	No namespace

Diagram	
Type	InstrumentInfoType
Properties	content: complex
Attributes	**QName** / **Type** / **Use** **Manufacturer** — xs:string — required for saving what columns were used **Name** — xs:string — optional **SerialNumber** — xs:string — optional for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.
Source	`<xs:element name="IDinfo" type="InstrumentInfoType"/>`

Element `InstrumentListType / ViscometerInfo`

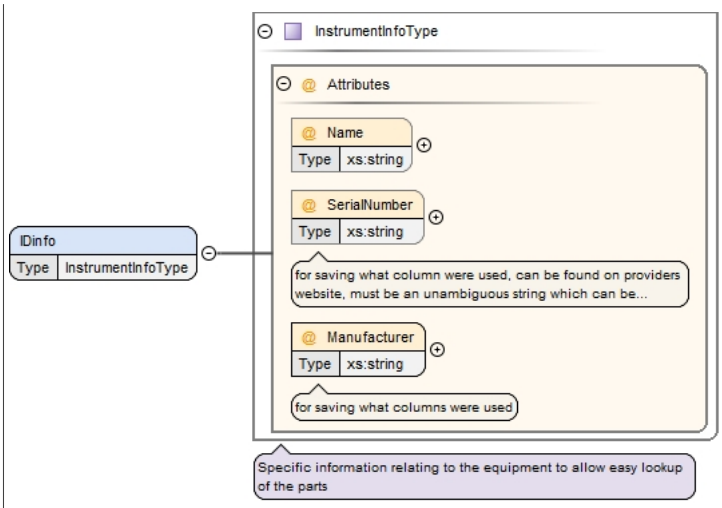
Namespace	No namespace
Diagram	
Type	ViscometerInfoType
Properties	content: complex minOccurs: 0
Model	IDinfo
Children	IDinfo

Instance	`<ViscometerInfo>`
	`  <IDinfo Manufacturer="" Name="" SerialNumber="">{1,1}</IDinfo>`
	`</ViscometerInfo>`

Source	`<xs:element name="ViscometerInfo" type="ViscometerInfoType" minOccurs="0"/>`

Element `ViscometerInfoType / IDinfo`

Namespace	No namespace

Diagram	
Type	InstrumentInfoType
Properties	content: complex

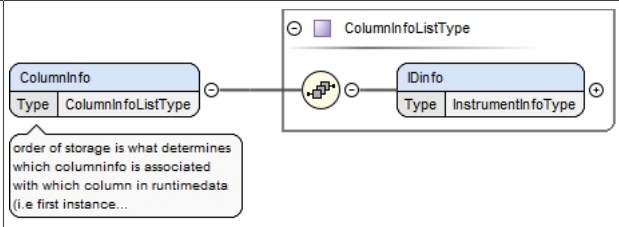
Attributes

QName	Type	Use	
Manufacturer	xs:string	required	
	for saving what columns were used		
Name	xs:string	optional	
SerialNumber	xs:string	optional	
	for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.		

Source

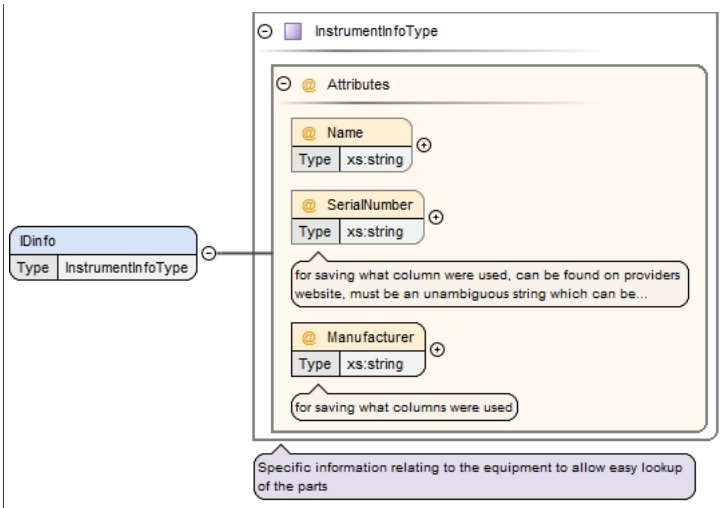
```
<xs:element name="IDinfo" type="InstrumentInfoType"/>
```

Element InstrumentListType / ColumnInfo

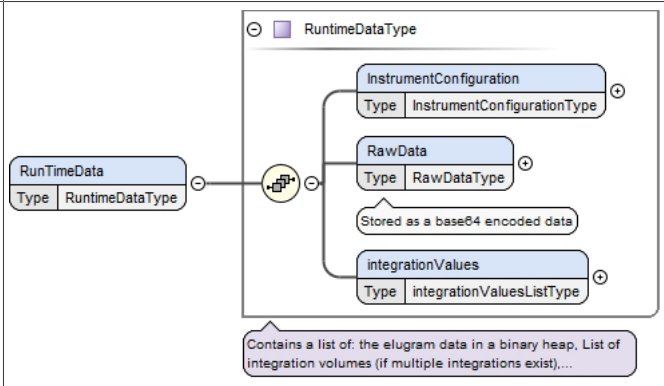
Namespace	No namespace
Annotations	order of storage is what determines which columninfo is associated with which column in runtimedata (i.e first instance of column info is related to the first instance of runtime data column
Diagram	
Type	ColumnInfoListType
Properties	content: complex
Model	IDinfo
Children	IDinfo
Instance	<ColumnInfo> <IDinfo Manufacturer="" Name="" SerialNumber="">{1,1}</IDinfo> </ColumnInfo>
Source	<xs:element name="ColumnInfo" type="ColumnInfoListType"> <xs:annotation> <xs:documentation>order of storage is what determines which columninfo is associated with which column in runtimedata (i.e first instance of column info is related to the first instance of runtime data column</xs:documentation> </xs:annotation> </xs:element>

Element ColumnInfoListType / IDinfo

Namespace	No namespace

Diagram	
Type	InstrumentInfoType
Properties	content: complex
Attributes	**QName** / **Type** / **Use** **Manufacturer** / xs:string / required for saving what columns were used **Name** / xs:string / optional **SerialNumber** / xs:string / optional for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.
Source	`<xs:element name="IDinfo" type="InstrumentInfoType"/>`

Element gpcMLType / RunTimeData

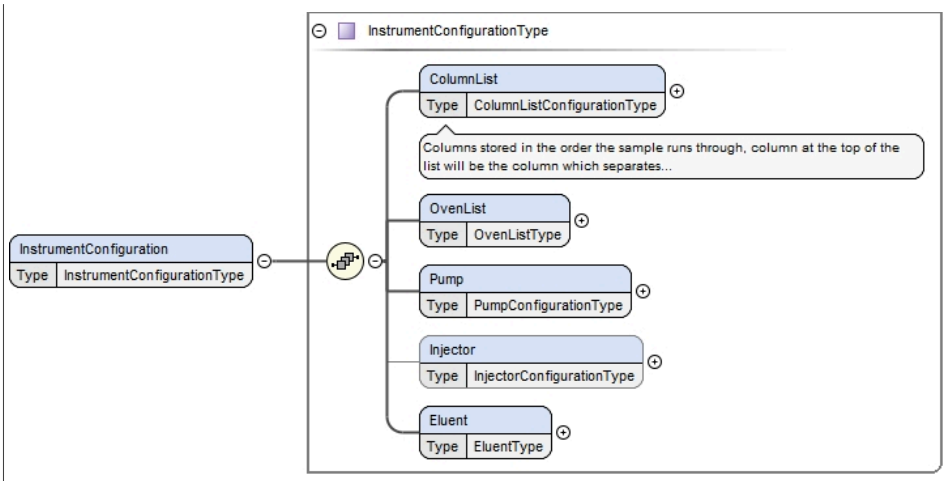
Namespace	No namespace
Diagram	
Type	RuntimeDataType
Properties	content: complex
Model	InstrumentConfiguration , RawData , integrationValues
Children	InstrumentConfiguration, RawData, integrationValues

Instance	`<RunTimeData>`
	`  <InstrumentConfiguration>{1,1}</InstrumentConfiguration>`
	`  <RawData Compression="">{1,1}</RawData>`
	`  <integrationValues>{1,1}</integrationValues>`
	`</RunTimeData>`

Source	`<xs:element name="RunTimeData" type="RuntimeDataType"/>`

Element RuntimeDataType / InstrumentConfiguration

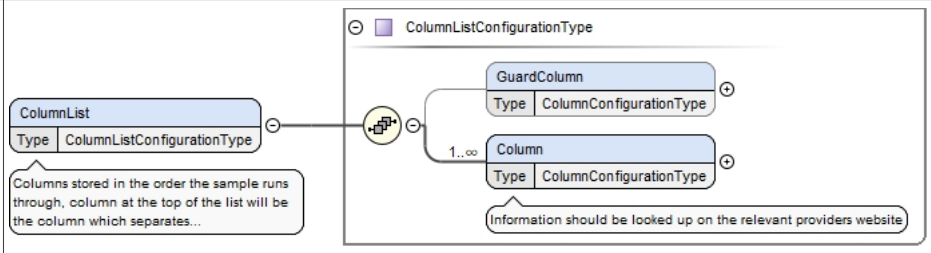
Namespace	No namespace

Diagram	
Type	InstrumentConfigurationType
Properties	content: complex
Model	ColumnList , OvenList , Pump , Injector{0,1} , Eluent
Children	ColumnList, Eluent, Injector, OvenList, Pump

Instance	`<InstrumentConfiguration>`
	`  <ColumnList>{1,1}</ColumnList>`
	`  <OvenList>{1,1}</OvenList>`
	`  <Pump FlowRate="">{1,1}</Pump>`
	`  <Injector InjectionVolume="">{0,1}</Injector>`
	`  <Eluent>{1,1}</Eluent>`
	`</InstrumentConfiguration>`

Source	`<xs:element name="InstrumentConfiguration" type="InstrumentConfigurationType"/>`

Element InstrumentConfigurationType / ColumnList

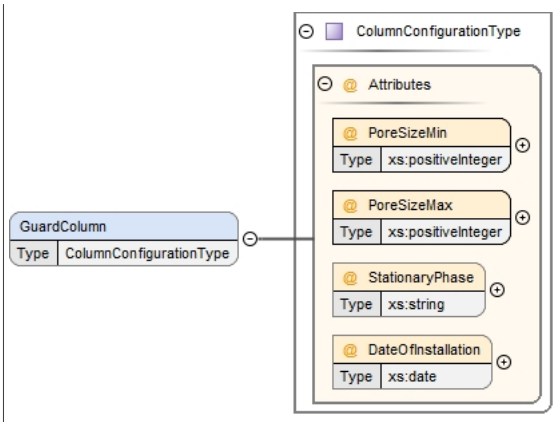
Namespace	No namespace
Annotations	Columns stored in the order the sample runs through, column at the top of the list will be the column which separates the sample first
Diagram	
Type	ColumnListConfigurationType
Properties	content: complex
Model	GuardColumn{0,1} , Column+
Children	Column, GuardColumn

Instance	`<ColumnList>`
	`  <GuardColumn DateOfInstallation="" PoreSizeMax="" PoreSizeMin="" StationaryPhase="">{0,1}</GuardColumn>`
	`  <Column DateOfInstallation="" PoreSizeMax="" PoreSizeMin="" StationaryPhase="">{1,unbounded}</Column>`
	`</ColumnList>`

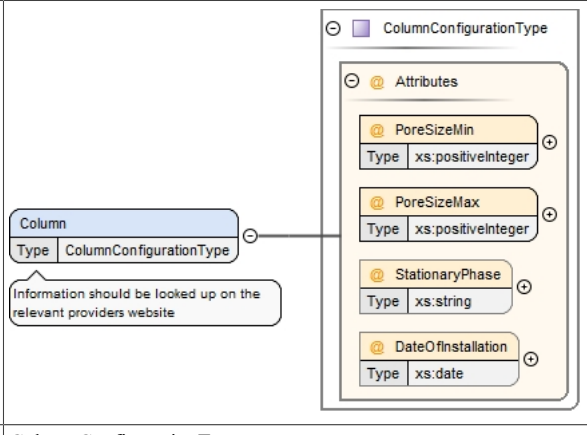
Source	`<xs:element name="ColumnList" type="ColumnListConfigurationType">`
	`  <xs:annotation>`
	`  <xs:documentation>Columns stored in the order the sample runs through, column at the top of the list will be the column which separates the sample first</xs:documentation>`
	`  </xs:annotation>`
	`</xs:element>`

Element ColumnListConfigurationType / GuardColumn

Namespace	No namespace

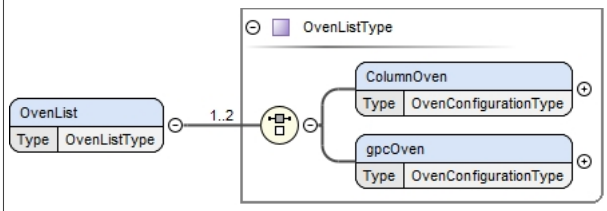
Diagram	
Type	ColumnConfigurationType
Properties	content: complex minOccurs: 0
Attributes	**QName** / **Type** / **Use** **DateOfInstallation** xs:date optional **PoreSizeMax** xs:positiveInteger required **PoreSizeMin** xs:positiveInteger required **StationaryPhase** xs:string optional
Source	`<xs:element name="GuardColumn" type="ColumnConfigurationType" minOccurs="0"/>`

Element `ColumnListConfigurationType / Column`

Namespace	No namespace
Annotations	`Information should be looked up on the relevant providers website`
Diagram	
Type	ColumnConfigurationType
Properties	content: complex maxOccurs: unbounded
Attributes	**QName** / **Type** / **Use** **DateOfInstallation** xs:date optional **PoreSizeMax** xs:positiveInteger required **PoreSizeMin** xs:positiveInteger required **StationaryPhase** xs:string optional

Source	`<xs:element name="Column" type="ColumnConfigurationType" maxOccurs="unbounded">`
	`  <xs:annotation>`
	`  <xs:documentation>Information should be looked up on the relevant providers website</xs:documentation>`
	`  </xs:annotation>`
	`</xs:element>`

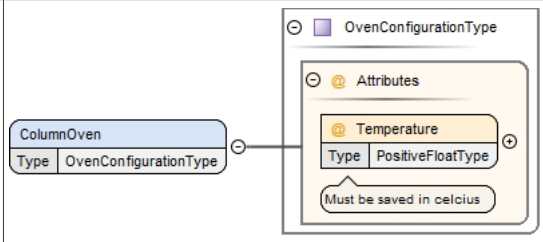
Element `InstrumentConfigurationType / OvenList`

Namespace	No namespace
Diagram	
Type	OvenListType
Properties	content: complex
Model	ColumnOven | gpcOven
Children	ColumnOven, gpcOven

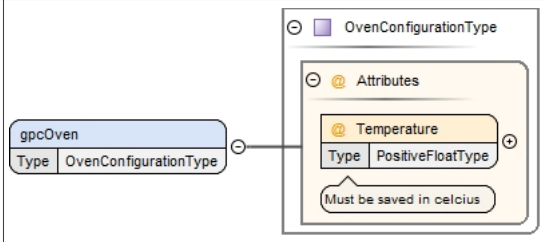
Instance	`<OvenList>`
	`  <ColumnOven Temperature="">{1,1}</ColumnOven>`
	`  <gpcOven Temperature="">{1,1}</gpcOven>`
	`</OvenList>`

Source	`<xs:element name="OvenList" type="OvenListType"/>`

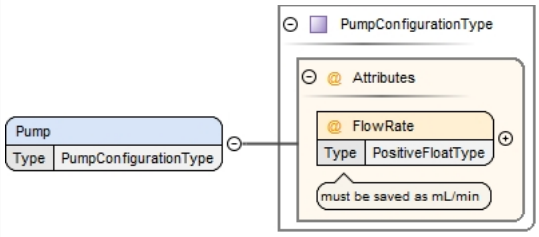
Element `OvenListType / ColumnOven`

Namespace	No namespace
Diagram	
Type	OvenConfigurationType
Properties	content: complex
Attributes	**QName** / **Type** / **Use** **Temperature** / PositiveFloatType / required Must be saved in celcius
Source	`<xs:element name="ColumnOven" type="OvenConfigurationType"/>`

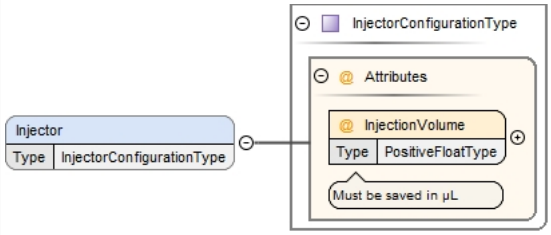
Element `OvenListType / gpcOven`

Namespace	No namespace
Diagram	
Type	OvenConfigurationType
Properties	content: complex
Attributes	**QName** / **Type** / **Use** **Temperature** / PositiveFloatType / required Must be saved in celcius
Source	`<xs:element name="gpcOven" type="OvenConfigurationType"/>`

Element `InstrumentConfigurationType / Pump`

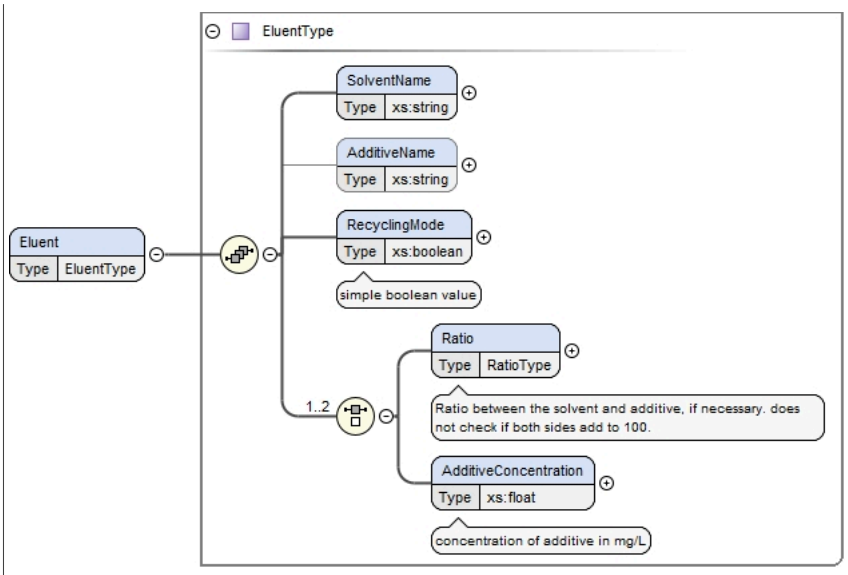
Namespace	No namespace
Diagram	
Type	PumpConfigurationType
Properties	content: `complex`
Attributes	**QName** / **Type** / **Use** **FlowRate** / PositiveFloatType / required `must be saved as mL/min`
Source	`<xs:element name="Pump" type="PumpConfigurationType"/>`

Element `InstrumentConfigurationType / Injector`

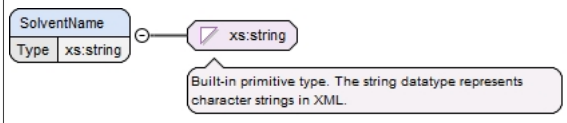
Namespace	No namespace
Diagram	
Type	InjectorConfigurationType
Properties	content: `complex` minOccurs: `0`
Attributes	**QName** / **Type** / **Use** **InjectionVolume** / PositiveFloatType / required `Must be saved in µL`
Source	`<xs:element name="Injector" type="InjectorConfigurationType" minOccurs="0"/>`

Element `InstrumentConfigurationType / Eluent`

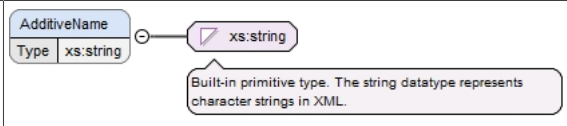
Namespace	No namespace

Diagram	
Type	EluentType
Properties	content: complex
Model	SolventName , AdditiveName{0,1} , RecyclingMode , (Ratio | AdditiveConcentration)
Children	AdditiveConcentration, AdditiveName, Ratio, RecyclingMode, SolventName
Instance	<Eluent> <SolventName>{1,1}</SolventName> <AdditiveName>{0,1}</AdditiveName> <RecyclingMode>{1,1}</RecyclingMode> <Ratio>{1,1}</Ratio> <AdditiveConcentration>{1,1}</AdditiveConcentration> </Eluent>
Source	<xs:element name="Eluent" type="EluentType"/>

Element EluentType / SolventName

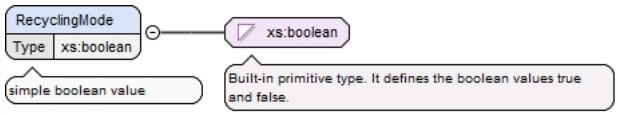
Namespace	No namespace
Diagram	
Type	xs:string
Properties	content: simple
Source	<xs:element name="SolventName" type="xs:string"/>

Element EluentType / AdditiveName

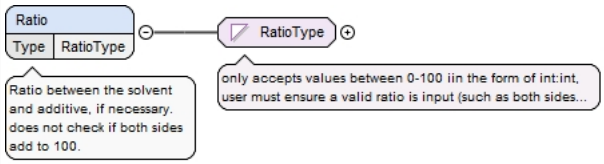
Namespace	No namespace
Diagram	
Type	xs:string
Properties	content: simple minOccurs: 0
Source	<xs:element name="AdditiveName" type="xs:string" minOccurs="0"/>

Element EluentType / RecyclingMode

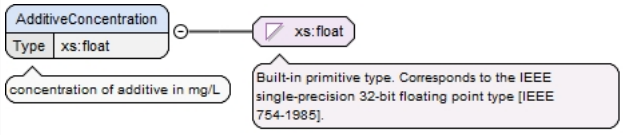
Namespace	No namespace

Annotations	simple boolean value
Diagram	
Type	xs:boolean
Properties	content: simple
Source	<xs:element name="RecyclingMode" type="xs:boolean"> <xs:annotation> <xs:documentation>simple boolean value</xs:documentation> </xs:annotation> </xs:element>

Element EluentType / Ratio

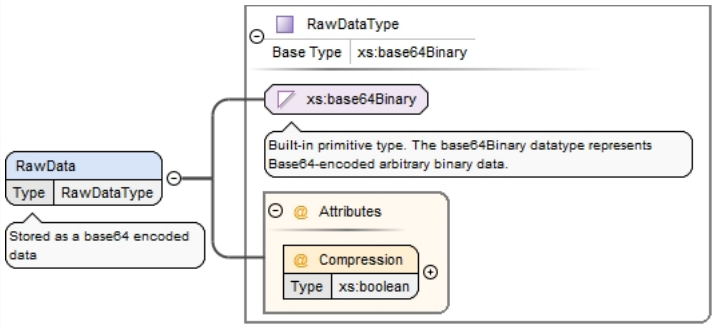
Namespace	No namespace
Annotations	Ratio between the solvent and additive, if necessary. does not check if both sides add to 100.
Diagram	
Type	RatioType
Properties	content: simple
Facets	pattern (100|[0-9]?[0-9]):(100|[0-9]?[0-9])
Source	<xs:element name="Ratio" type="RatioType"> <xs:annotation> <xs:documentation>Ratio between the solvent and additive, if necessary. does not check if both sides add to 100.</xs:documentation> </xs:annotation> </xs:element>

Element EluentType / AdditiveConcentration

Namespace	No namespace
Annotations	concentration of additive in mg/L
Diagram	
Type	xs:float
Properties	content: simple
Source	<xs:element name="AdditiveConcentration" type="xs:float"> <xs:annotation> <xs:documentation>concentration of additive in mg/L</xs:documentation> </xs:annotation> </xs:element>

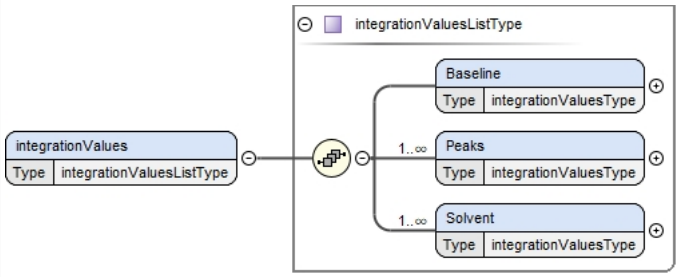
Element RuntimeDataType / RawData

Namespace	No namespace
Annotations	Stored as a base64 encoded data

Diagram	
Type	RawDataType
Properties	content: complex
Attributes	**QName** / **Type** / **Use** **Compression** / xs:boolean / required

Source	`<xs:element name="RawData" type="RawDataType">`
	`  <xs:annotation>`
	`  <xs:documentation>Stored as a base64 encoded data</xs:documentation>`
	`  </xs:annotation>`
	`</xs:element>`

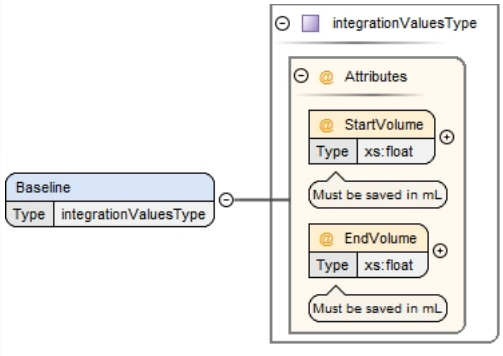
Element RuntimeDataType / integrationValues

Namespace	No namespace
Diagram	
Type	integrationValuesListType
Properties	content: complex
Model	Baseline , Peaks+ , Solvent+
Children	Baseline, Peaks, Solvent

Instance	`<integrationValues>`
	`  <Baseline EndVolume="" StartVolume="">{1,1}</Baseline>`
	`  <Peaks EndVolume="" StartVolume="">{1,unbounded}</Peaks>`
	`  <Solvent EndVolume="" StartVolume="">{1,unbounded}</Solvent>`
	`</integrationValues>`

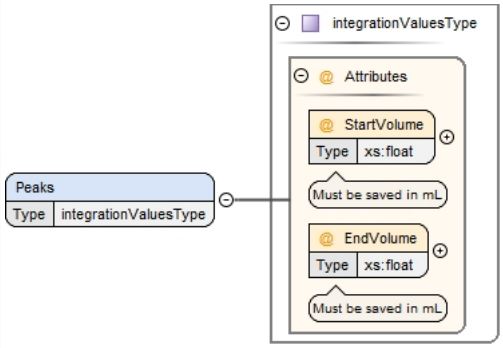
Source	`<xs:element name="integrationValues" type="integrationValuesListType"/>`

Element integrationValuesListType / Baseline

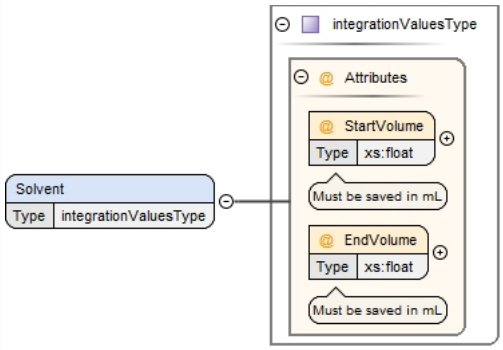
Namespace	No namespace
Diagram	

Type	integrationValuesType
Properties	content: complex
Attributes	**QName** / **Type** / **Use** **EndVolume** / xs:float / required Must be saved in mL **StartVolume** / xs:float / required Must be saved in mL
Source	`<xs:element name="Baseline" type="integrationValuesType"/>`

Element integrationValuesListType / Peaks

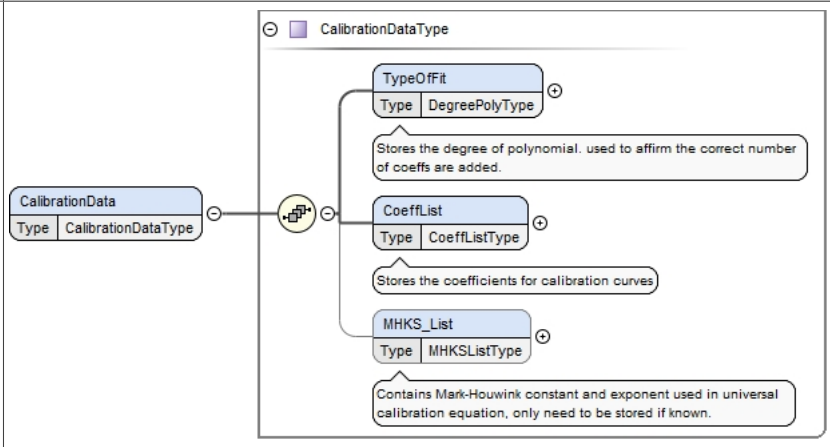
Namespace	No namespace
Diagram	
Type	integrationValuesType
Properties	content: complex maxOccurs: unbounded
Attributes	**QName** / **Type** / **Use** **EndVolume** / xs:float / required Must be saved in mL **StartVolume** / xs:float / required Must be saved in mL
Source	`<xs:element name="Peaks" type="integrationValuesType" maxOccurs="unbounded"/>`

Element integrationValuesListType / Solvent

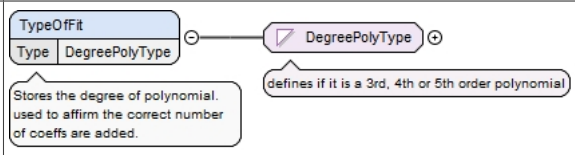
Namespace	No namespace
Diagram	
Type	integrationValuesType
Properties	content: complex maxOccurs: unbounded
Attributes	**QName** / **Type** / **Use** **EndVolume** / xs:float / required Must be saved in mL

	QName	Type	Use	
	StartVolume	xs:float	required	
		Must be saved in mL		
Source	<xs:element name="Solvent" type="integrationValuesType" maxOccurs="unbounded"/>			

Element gpcMLType / CalibrationData

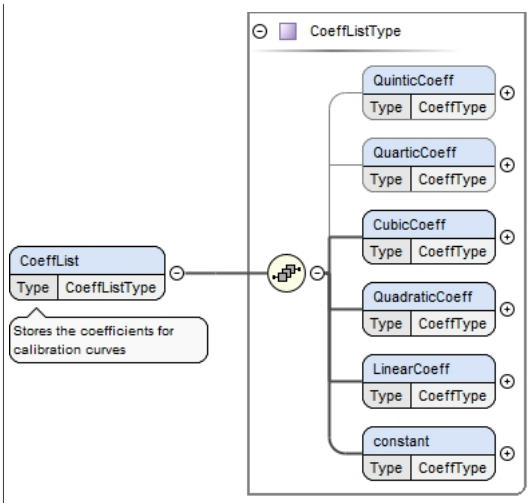
Namespace	No namespace
Diagram	
Type	CalibrationDataType
Properties	content: complex
Model	TypeOfFit , CoeffList , MHKS_List{0,1}
Children	CoeffList, MHKS_List, TypeOfFit
Instance	<CalibrationData> <TypeOfFit>{1,1}</TypeOfFit> <CoeffList>{1,1}</CoeffList> <MHKS_List>{0,1}</MHKS_List> </CalibrationData>
Source	<xs:element name="CalibrationData" type="CalibrationDataType"/>

Element CalibrationDataType / TypeOfFit

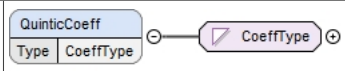
Namespace	No namespace
Annotations	Stores the degree of polynomial. used to affirm the correct number of coeffs are added.
Diagram	
Type	DegreePolyType
Properties	content: simple
Facets	maxInclusive 5 minInclusive 3
Source	<xs:element name="TypeOfFit" type="DegreePolyType"> <xs:annotation> <xs:documentation>Stores the degree of polynomial. used to affirm the correct number of coeffs are added.</xs:documentation> </xs:annotation> </xs:element>

Element CalibrationDataType / CoeffList

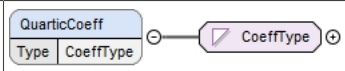
Namespace	No namespace
Annotations	Stores the coefficients for calibration curves

Diagram	
Type	CoeffListType
Properties	content: complex
Model	QuinticCoeff{0,1} , QuarticCoeff{0,1} , CubicCoeff , QuadraticCoeff , LinearCoeff , constant
Children	CubicCoeff, LinearCoeff, QuadraticCoeff, QuarticCoeff, QuinticCoeff, constant
Instance	<pre><CoeffList> <QuinticCoeff>{0,1}</QuinticCoeff> <QuarticCoeff>{0,1}</QuarticCoeff> <CubicCoeff>{1,1}</CubicCoeff> <QuadraticCoeff>{1,1}</QuadraticCoeff> <LinearCoeff>{1,1}</LinearCoeff> <constant>{1,1}</constant> </CoeffList></pre>
Source	<pre><xs:element name="CoeffList" type="CoeffListType"> <xs:annotation> <xs:documentation>Stores the coefficients for calibration curves</xs:documentation> </xs:annotation> </xs:element></pre>

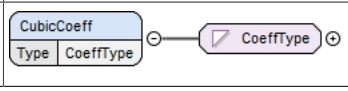
Element CoeffListType / QuinticCoeff

Namespace	No namespace
Diagram	
Type	CoeffType
Properties	content: simple minOccurs: 0
Source	<pre><xs:element name="QuinticCoeff" type="CoeffType" minOccurs="0"/></pre>

Element CoeffListType / QuarticCoeff

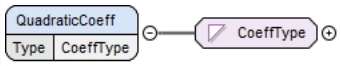
Namespace	No namespace
Diagram	
Type	CoeffType
Properties	content: simple minOccurs: 0
Source	<pre><xs:element name="QuarticCoeff" type="CoeffType" minOccurs="0"/></pre>

Element CoeffListType / CubicCoeff

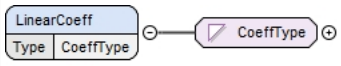
Namespace	No namespace
Diagram	

Type	CoeffType
Properties	content: simple
Source	`<xs:element name="CubicCoeff" type="CoeffType"/>`

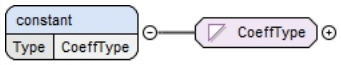
Element CoeffListType / QuadraticCoeff

Namespace	No namespace
Diagram	
Type	CoeffType
Properties	content: simple
Source	`<xs:element name="QuadraticCoeff" type="CoeffType"/>`

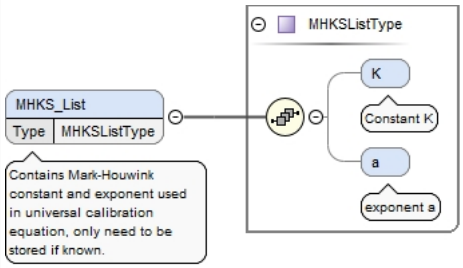
Element CoeffListType / LinearCoeff

Namespace	No namespace
Diagram	
Type	CoeffType
Properties	content: simple
Source	`<xs:element name="LinearCoeff" type="CoeffType"/>`

Element CoeffListType / constant

Namespace	No namespace
Diagram	
Type	CoeffType
Properties	content: simple
Source	`<xs:element name="constant" type="CoeffType"/>`

Element CalibrationDataType / MHKS_List

Namespace	No namespace
Annotations	Contains Mark-Houwink constant and exponent used in universal calibration equation, only need to be stored if known.
Diagram	
Type	MHKSListType
Properties	content: complex minOccurs: 0
Model	K{0,1} , a{0,1}
Children	K, a

Instance	`<MHKS_List>`
	`  <K>{0,1}</K>`
	`  <a>{0,1}</a>`
	`</MHKS_List>`

Source	`<xs:element name="MHKS_List" type="MHKSListType" minOccurs="0">` `  <xs:annotation>`

```
    <xs:documentation>Contains Mark-Houwink constant and exponent used in universal calibration
 equation, only need to be stored if known.</xs:documentation>
  </xs:annotation>
</xs:element>
```

Element MHKSListType / K

Namespace	No namespace
Annotations	Constant K
Diagram	
Properties	minOccurs: 0

Source	`<xs:element name="K" minOccurs="0">`
	`  <xs:annotation>`
	`  <xs:documentation>Constant K</xs:documentation>`
	`  </xs:annotation>`
	`</xs:element>`

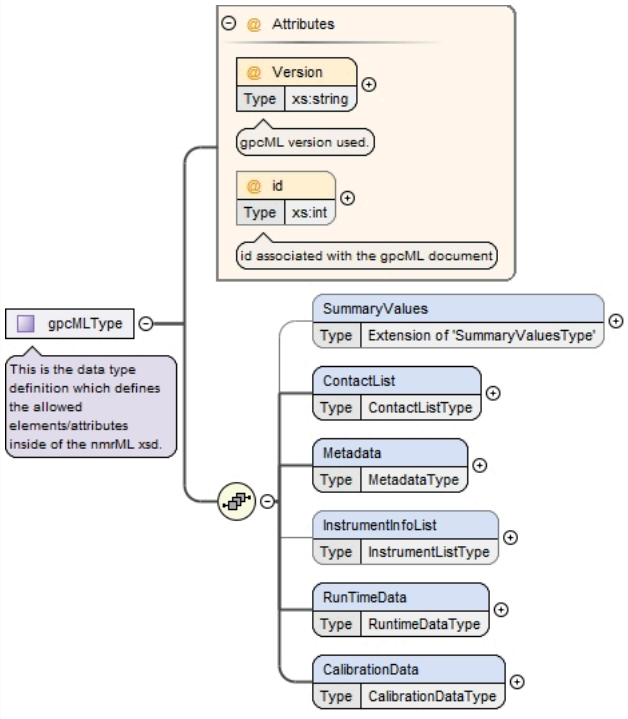
Element MHKSListType / a

Namespace	No namespace
Annotations	exponent a
Diagram	
Properties	minOccurs: 0

Source	`<xs:element name="a" minOccurs="0">`
	`  <xs:annotation>`
	`  <xs:documentation>exponent a</xs:documentation>`
	`  </xs:annotation>`
	`</xs:element>`

Complex Type(s)

Complex Type gpcMLType

Namespace	No namespace
Annotations	This is the data type definition which defines the allowed elements/attributes inside of the nmrML xsd.
Diagram	

Used by	Element gpcML
Model	SummaryValues{0,1} , ContactList , Metadata , InstrumentInfoList{0,1} , RunTimeData , CalibrationData
Children	CalibrationData, ContactList, InstrumentInfoList, Metadata, RunTimeData, SummaryValues

Attributes

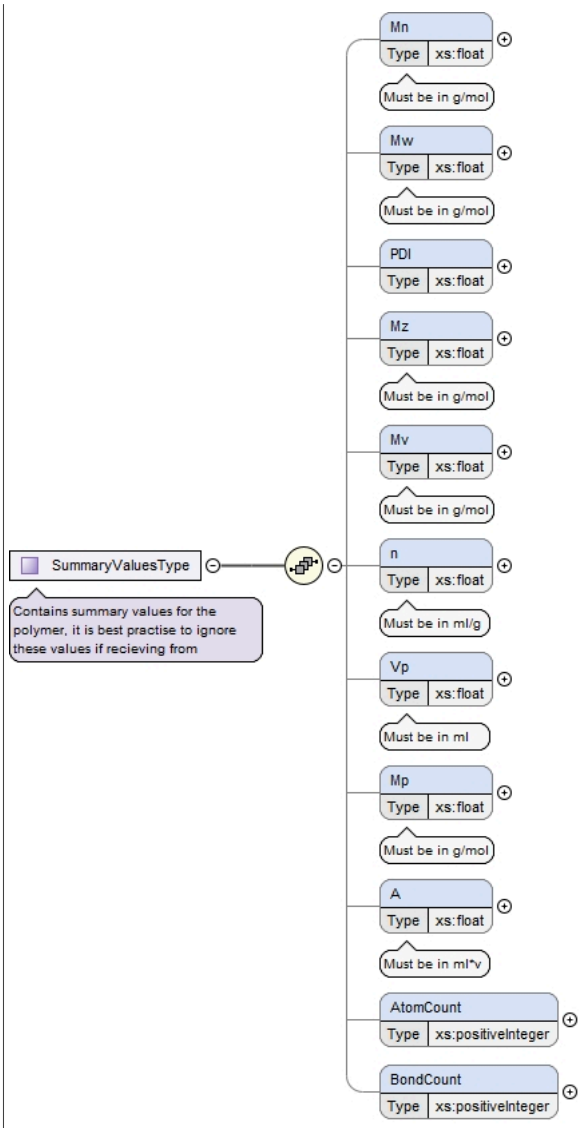
QName	Type	Use	
Version	xs:string	required	
	gpcML version used.		
id	xs:int	optional	
	id associated with the gpcML document		

Source

```
<xs:complexType name="gpcMLType">
  <xs:annotation>
    <xs:documentation>This is the data type definition which defines the allowed elements/attributes
 inside of the nmrML xsd.</xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="SummaryValues" minOccurs="0">
      <xs:complexType>
        <xs:complexContent>
          <xs:extension base="SummaryValuesType"/>
        </xs:complexContent>
      </xs:complexType>
    </xs:element>
    <xs:element name="ContactList" type="ContactListType"/>
    <xs:element name="Metadata" type="MetadataType"/>
    <xs:element name="InstrumentInfoList" type="InstrumentListType" minOccurs="0"/>
    <xs:element name="RunTimeData" type="RuntimeDataType"/>
    <xs:element name="CalibrationData" type="CalibrationDataType"/>
  </xs:sequence>
  <xs:attribute name="Version" type="xs:string" use="required">
    <xs:annotation>
      <xs:documentation>gpcML version used.</xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="id" type="xs:int">
    <xs:annotation>
      <xs:documentation>id associated with the gpcML document</xs:documentation>
    </xs:annotation>
  </xs:attribute>
</xs:complexType>
```

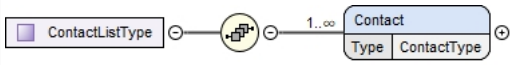
Complex Type SummaryValuesType

Namespace	No namespace
Annotations	Contains summary values for the polymer, it is best practise to ignore these values if recieving from

Diagram	
Used by	Element gpcMLType/SummaryValues
Model	Mn{0,1} , Mw{0,1} , PDI{0,1} , Mz{0,1} , Mv{0,1} , n{0,1} , Vp{0,1} , Mp{0,1} , A{0,1} , AtomCount{0,1} , BondCount{0,1}
Children	A, AtomCount, BondCount, Mn, Mp, Mv, Mw, Mz, PDI, Vp, n
Source	<xs:complexType name="SummaryValuesType"> <xs:annotation> <xs:documentation>Contains summary values for the polymer, it is best practise to ignore these values if recieving from</xs:documentation> </xs:annotation> <xs:sequence> <xs:element name="Mn" minOccurs="0" type="xs:float"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element> <xs:element name="Mw" minOccurs="0" type="xs:float"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element> <xs:element name="PDI" minOccurs="0" type="xs:float"/> <xs:element name="Mz" minOccurs="0" type="xs:float"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element> <xs:element name="Mv" minOccurs="0" type="xs:float"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation>

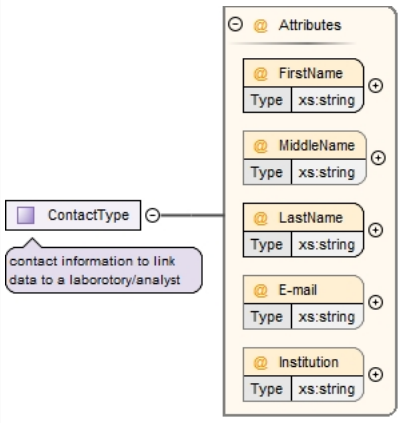
```
    </xs:element>
    <xs:element name="n" type="xs:float" minOccurs="0">
      <xs:annotation>
        <xs:documentation>Must be in ml/g</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="Vp" minOccurs="0" type="xs:float">
      <xs:annotation>
        <xs:documentation>Must be in ml</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="Mp" minOccurs="0" type="xs:float">
      <xs:annotation>
        <xs:documentation>Must be in g/mol</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="A" minOccurs="0" type="xs:float">
      <xs:annotation>
        <xs:documentation>Must be in ml*v</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element minOccurs="0" name="AtomCount" type="xs:positiveInteger"/>
    <xs:element minOccurs="0" name="BondCount" type="xs:positiveInteger"/>
  </xs:sequence>
</xs:complexType>
```

Complex Type `ContactListType`

Namespace	No namespace
Diagram	
Used by	Element gpcMLType/ContactList
Model	Contact+
Children	Contact

Source	`<xs:complexType name="ContactListType">`
	`  <xs:sequence>`
	`  <xs:element name="Contact" type="ContactType" maxOccurs="unbounded"/>`
	`  </xs:sequence>`
	`</xs:complexType>`

Complex Type `ContactType`

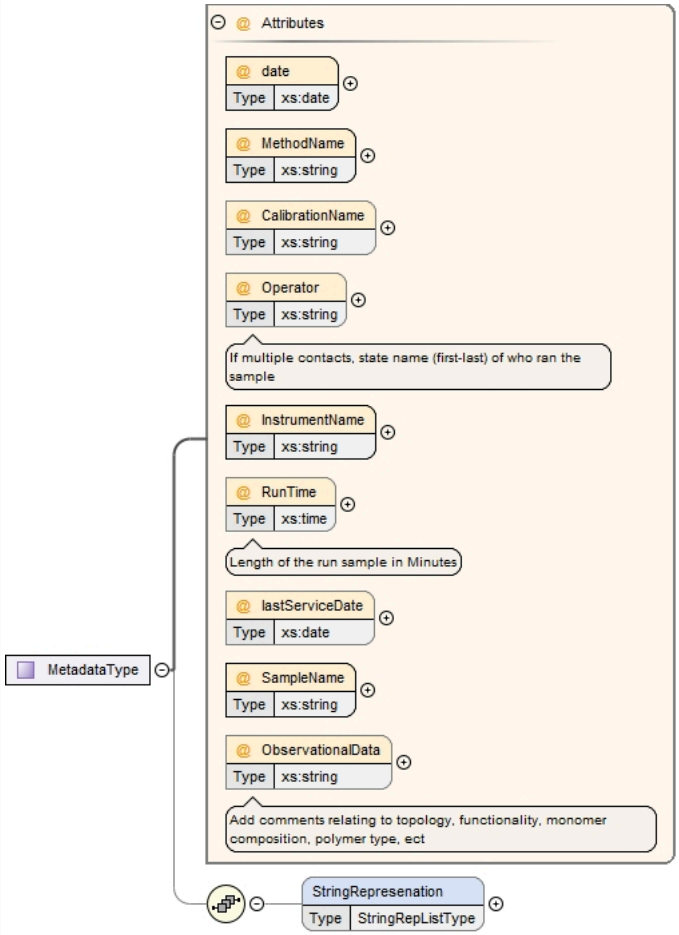
Namespace	No namespace
Annotations	contact information to link data to a laborotory/analyst
Diagram	
Used by	Element ContactListType/Contact

Attributes

QName	Type	Use	
E-mail	xs:string	optional	
FirstName	xs:string	required	
Institution	xs:string	optional	
LastName	xs:string	required	
MiddleName	xs:string	optional	

Source	<xs:complexType name="ContactType"> <xs:annotation> <xs:documentation>contact information to link data to a laborotory/analyst</xs:documentation> </xs:annotation> <xs:attribute name="FirstName" type="xs:string" use="required"/> <xs:attribute name="MiddleName" type="xs:string" use="optional"/> <xs:attribute name="LastName" type="xs:string" use="required"/> <xs:attribute name="E-mail" type="xs:string"/> <xs:attribute name="Institution" type="xs:string" use="optional"/> </xs:complexType>

Complex Type `MetadataType`

Namespace	No namespace
Diagram	
Used by	Element gpcMLType/Metadata
Model	StringRepresenation{0,1}
Children	StringRepresenation

Attributes

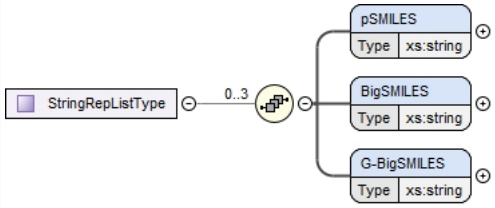
QName	Type	Use	
CalibrationName	xs:string	optional	
InstrumentName	xs:string	required	
MethodName	xs:string	required	
ObservationalData	xs:string	optional	
	`Add comments relating to topology, functionality, monomer composition, polymer type, ect`		
Operator	xs:string	optional	
	`If multiple contacts, state name (first-last) of who ran the sample`		
RunTime	xs:time	optional	
	`Length of the run sample in Minutes`		
SampleName	xs:string	required	

	QName	Type	Use	
	date	xs:date	required	
	lastServiceDate	xs:date	optional	

Source

```
<xs:complexType name="MetadataType">
  <xs:sequence minOccurs="0">
    <xs:element minOccurs="0" name="StringRepresenation" type="StringRepListType"/>
  </xs:sequence>
  <xs:attribute name="date" type="xs:date" use="required"/>
  <xs:attribute name="MethodName" type="xs:string" use="required"/>
  <xs:attribute name="CalibrationName" type="xs:string" use="optional"/>
  <xs:attribute name="Operator" type="xs:string">
    <xs:annotation>
      <xs:documentation>If multiple contacts, state name (first-last) of who ran the sample</
xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="InstrumentName" type="xs:string" use="required"/>
  <xs:attribute name="RunTime" type="xs:time" use="optional">
    <xs:annotation>
      <xs:documentation>Length of the run sample in Minutes</xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="lastServiceDate" type="xs:date" use="optional"/>
  <xs:attribute name="SampleName" type="xs:string" use="required"/>
  <xs:attribute name="ObservationalData" type="xs:string">
    <xs:annotation>
      <xs:documentation>Add comments relating to topology, functionality, monomer composition,
 polymer type, ect</xs:documentation>
    </xs:annotation>
  </xs:attribute>
</xs:complexType>
```

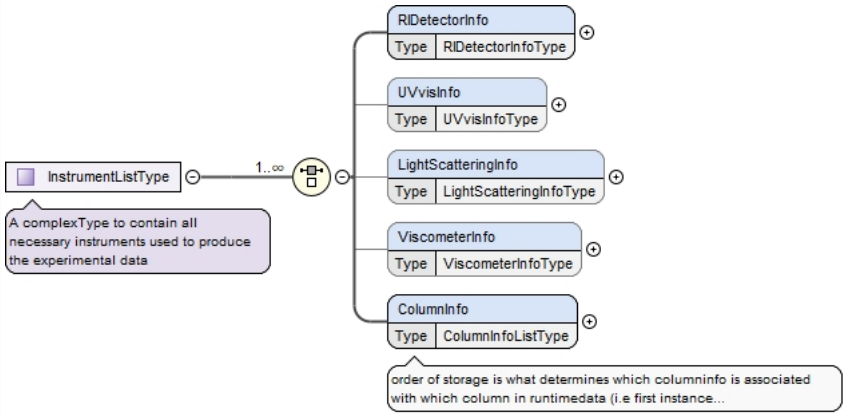
Complex Type `StringRepListType`

Namespace	No namespace
Diagram	
Used by	Element MetadataType/StringRepresenation
Model	pSMILES , BigSMILES , G-BigSMILES
Children	BigSMILES, G-BigSMILES, pSMILES
Source	


```
<xs:complexType name="StringRepListType">
  <xs:sequence maxOccurs="3" minOccurs="0">
    <xs:element name="pSMILES" type="xs:string"/>
    <xs:element name="BigSMILES" type="xs:string"/>
    <xs:element name="G-BigSMILES" type="xs:string"/>
  </xs:sequence>
</xs:complexType>
```

Complex Type `InstrumentListType`

Namespace	No namespace
Annotations	A complexType to contain all necessary instruments used to produce the experimental data

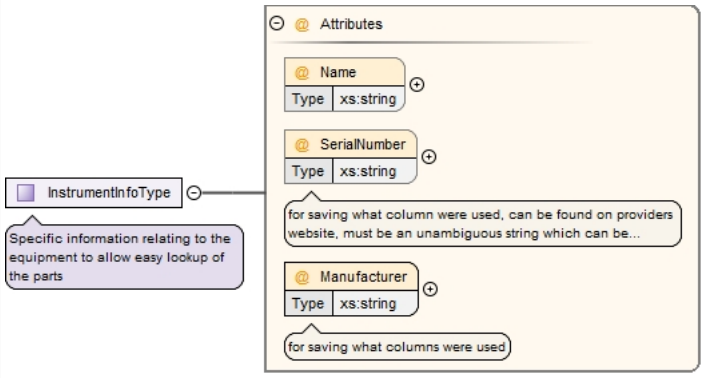
Diagram	
Used by	Element gpcMLType/InstrumentInfoList
Model	RIDetectorInfo | UVvisInfo{0,1} | LightScatteringInfo{0,1} | ViscometerInfo{0,1} | ColumnInfo
Children	ColumnInfo, LightScatteringInfo, RIDetectorInfo, UVvisInfo, ViscometerInfo
Source	<xs:complexType name="InstrumentListType"> <xs:annotation> <xs:documentation>A complexType to contain all necessary instruments used to produce the experimental data</xs:documentation> </xs:annotation> <xs:choice maxOccurs="unbounded"> <xs:element name="RIDetectorInfo" type="RIDetectorInfoType"/> <xs:element name="UVvisInfo" type="UVvisInfoType" minOccurs="0"/> <xs:element name="LightScatteringInfo" type="LightScatteringInfoType" minOccurs="0"/> <xs:element name="ViscometerInfo" type="ViscometerInfoType" minOccurs="0"/> <xs:element name="ColumnInfo" type="ColumnInfoListType"> <xs:annotation> <xs:documentation>order of storage is what determines which columninfo is associated with which column in runtimedata (i.e first instance of column info is related to the first instance of runtime data column</xs:documentation> </xs:annotation> </xs:element> </xs:choice> </xs:complexType>

Complex Type `RIDetectorInfoType`

Namespace	No namespace
Diagram	
Used by	Element InstrumentListType/RIDetectorInfo
Model	IDinfo
Children	IDinfo
Source	<xs:complexType name="RIDetectorInfoType"> <xs:sequence> <xs:element name="IDinfo" type="InstrumentInfoType"/> </xs:sequence> </xs:complexType>

Complex Type `InstrumentInfoType`

Namespace	No namespace
Annotations	Specific information relating to the equipment to allow easy lookup of the parts

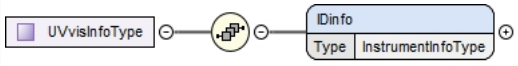
Diagram	
Used by	Elements ColumnInfoListType/IDinfo, LightScatteringInfoType/IDinfo, RIDetectorInfoType/IDinfo, UVvisInfoType/IDinfo, ViscometerInfoType/IDinfo

Attributes	QName	Type	Use	
	Manufacturer	xs:string	required	
		for saving what columns were used		
	Name	xs:string	optional	
	SerialNumber	xs:string	optional	
		for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.		

Source

```
<xs:complexType name="InstrumentInfoType">
  <xs:annotation>
    <xs:documentation>Specific information relating to the equipment to allow easy lookup of the
 parts</xs:documentation>
  </xs:annotation>
  <xs:attribute name="Name" type="xs:string" use="optional"/>
  <xs:attribute name="SerialNumber" type="xs:string">
    <xs:annotation>
      <xs:documentation>for saving what column were used, can be found on providers website, must be
 an unambiguous string which can be directly linked to the item.</xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="Manufacturer" type="xs:string" use="required">
    <xs:annotation>
      <xs:documentation>for saving what columns were used</xs:documentation>
    </xs:annotation>
  </xs:attribute>
</xs:complexType>
```

Complex Type `UVvisInfoType`

Namespace	No namespace
Diagram	
Used by	Element InstrumentListType/UVvisInfo
Model	IDinfo
Children	IDinfo

Source

```
<xs:complexType name="UVvisInfoType">
  <xs:sequence>
    <xs:element name="IDinfo" type="InstrumentInfoType"/>
  </xs:sequence>
</xs:complexType>
```

Complex Type `LightScatteringInfoType`

Namespace	No namespace
Diagram	
Used by	Element InstrumentListType/LightScatteringInfo

Model	IDinfo
Children	IDinfo
Source	<xs:complexType name="LightScatteringInfoType"> <xs:sequence> <xs:element name="IDinfo" type="InstrumentInfoType"/> </xs:sequence> </xs:complexType>

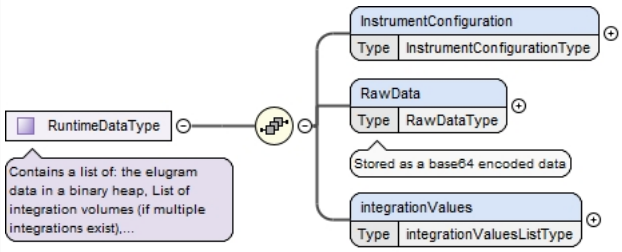
Complex Type `ViscometerInfoType`

Namespace	No namespace
Diagram	
Used by	Element InstrumentListType/ViscometerInfo
Model	IDinfo
Children	IDinfo
Source	<xs:complexType name="ViscometerInfoType"> <xs:sequence> <xs:element name="IDinfo" type="InstrumentInfoType"/> </xs:sequence> </xs:complexType>

Complex Type `ColumnInfoListType`

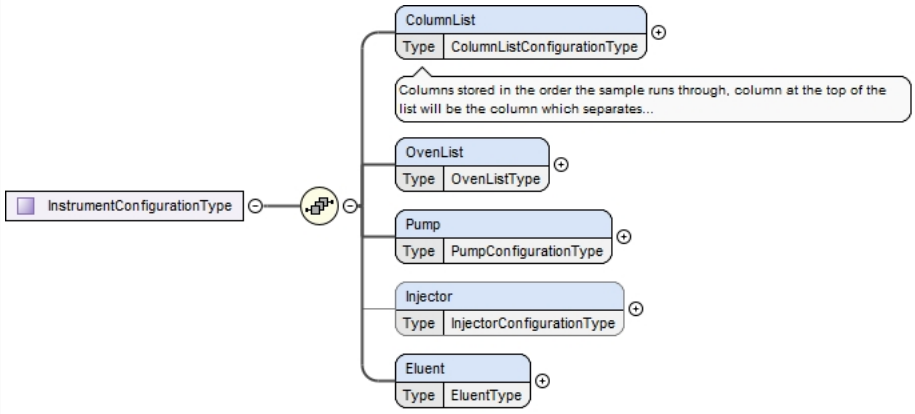
Namespace	No namespace
Diagram	
Used by	Element InstrumentListType/ColumnInfo
Model	IDinfo
Children	IDinfo
Source	<xs:complexType name="ColumnInfoListType"> <xs:sequence> <xs:element name="IDinfo" type="InstrumentInfoType"/> </xs:sequence> </xs:complexType>

Complex Type `RuntimeDataType`

Namespace	No namespace
Annotations	Contains a list of: the elugram data in a binary heap, List of integration volumes (if multiple integrations exist), system peak and internal standard peak volumes
Diagram	
Used by	Element gpcMLType/RunTimeData
Model	InstrumentConfiguration , RawData , integrationValues
Children	InstrumentConfiguration, RawData, integrationValues
Source	<xs:complexType name="RuntimeDataType"> <xs:annotation> <xs:documentation>Contains a list of: the elugram data in a binary heap, List of integration volumes (if multiple integrations exist), system peak and internal standard peak volumes</xs:documentation> </xs:annotation>

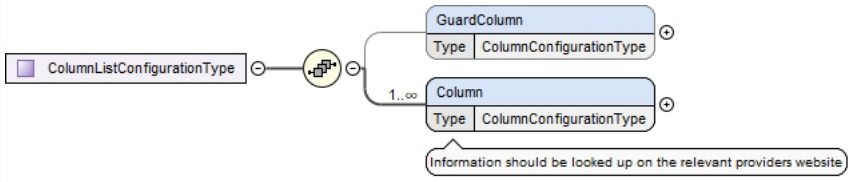
```
  <xs:sequence>
    <xs:element name="InstrumentConfiguration" type="InstrumentConfigurationType"/>
    <xs:element name="RawData" type="RawDataType">
      <xs:annotation>
        <xs:documentation>Stored as a base64 encoded data</xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="integrationValues" type="integrationValuesListType"/>
  </xs:sequence>
</xs:complexType>
```

Complex Type `InstrumentConfigurationType`

Namespace	No namespace
Diagram	
Used by	Element RuntimeDataType/InstrumentConfiguration
Model	ColumnList , OvenList , Pump , Injector{0,1} , Eluent
Children	ColumnList, Eluent, Injector, OvenList, Pump

Source	
	`<xs:complexType name="InstrumentConfigurationType">`
	`  <xs:sequence>`
	`    <xs:element name="ColumnList" type="ColumnListConfigurationType">`
	`      <xs:annotation>`
	`        <xs:documentation>Columns stored in the order the sample runs through, column at the top of the list will be the column which separates the sample first</xs:documentation>`
	`      </xs:annotation>`
	`    </xs:element>`
	`    <xs:element name="OvenList" type="OvenListType"/>`
	`    <xs:element name="Pump" type="PumpConfigurationType"/>`
	`    <xs:element name="Injector" type="InjectorConfigurationType" minOccurs="0"/>`
	`    <xs:element name="Eluent" type="EluentType"/>`
	`  </xs:sequence>`
	`</xs:complexType>`

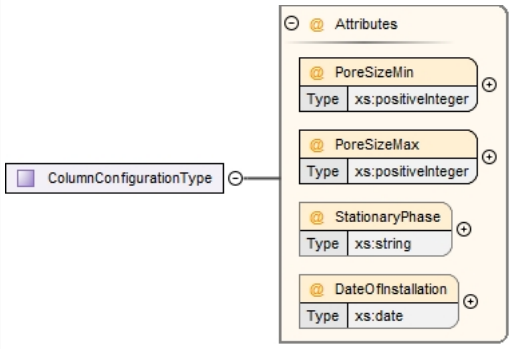
Complex Type `ColumnListConfigurationType`

Namespace	No namespace
Diagram	
Used by	Element InstrumentConfigurationType/ColumnList
Model	GuardColumn{0,1} , Column+
Children	Column, GuardColumn

Source	
	`<xs:complexType name="ColumnListConfigurationType">`
	`  <xs:sequence>`
	`    <xs:element name="GuardColumn" type="ColumnConfigurationType" minOccurs="0"/>`
	`    <xs:element name="Column" type="ColumnConfigurationType" maxOccurs="unbounded">`
	`      <xs:annotation>`
	`        <xs:documentation>Information should be looked up on the relevant providers website</xs:documentation>`

```
      </xs:annotation>
    </xs:element>
  </xs:sequence>
</xs:complexType>
```

Complex Type `ColumnConfigurationType`

Namespace	No namespace
Diagram	
Used by	Elements ColumnListConfigurationType/Column, ColumnListConfigurationType/GuardColumn

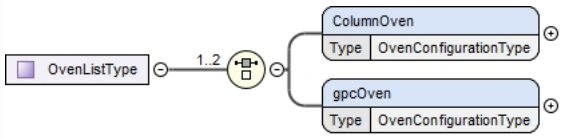
Attributes

QName	Type	Use	
DateOfInstallation	xs:date	optional	
PoreSizeMax	xs:positiveInteger	required	
PoreSizeMin	xs:positiveInteger	required	
StationaryPhase	xs:string	optional	

Source

```
<xs:complexType name="ColumnConfigurationType">
  <xs:attribute name="PoreSizeMin" type="xs:positiveInteger" use="required"/>
  <xs:attribute name="PoreSizeMax" type="xs:positiveInteger" use="required"/>
  <xs:attribute name="StationaryPhase" type="xs:string" use="optional"/>
  <xs:attribute name="DateOfInstallation" type="xs:date" use="optional"/>
</xs:complexType>
```

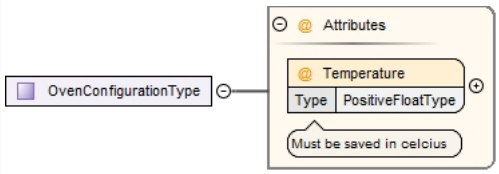
Complex Type `OvenListType`

Namespace	No namespace
Diagram	
Used by	Element InstrumentConfigurationType/OvenList
Model	ColumnOven | gpcOven
Children	ColumnOven, gpcOven

Source

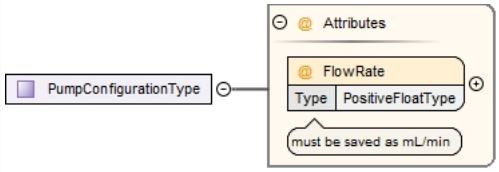
```
<xs:complexType name="OvenListType">
  <xs:choice maxOccurs="2">
    <xs:element name="ColumnOven" type="OvenConfigurationType"/>
    <xs:element name="gpcOven" type="OvenConfigurationType"/>
  </xs:choice>
</xs:complexType>
```

Complex Type `OvenConfigurationType`

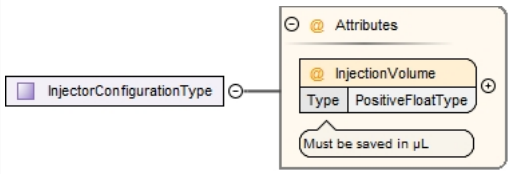
Namespace	No namespace
Diagram	

Used by	Elements	OvenListType/ColumnOven, OvenListType/gpcOven		
Attributes	**QName**	**Type**	**Use**	
	Temperature	PositiveFloatType	required	
		Must be saved in celcius		
Source	<xs:complexType name="OvenConfigurationType"> <xs:attribute name="Temperature" type="PositiveFloatType" use="required"> <xs:annotation> <xs:documentation>Must be saved in celcius</xs:documentation> </xs:annotation> </xs:attribute> </xs:complexType>			

Complex Type `PumpConfigurationType`

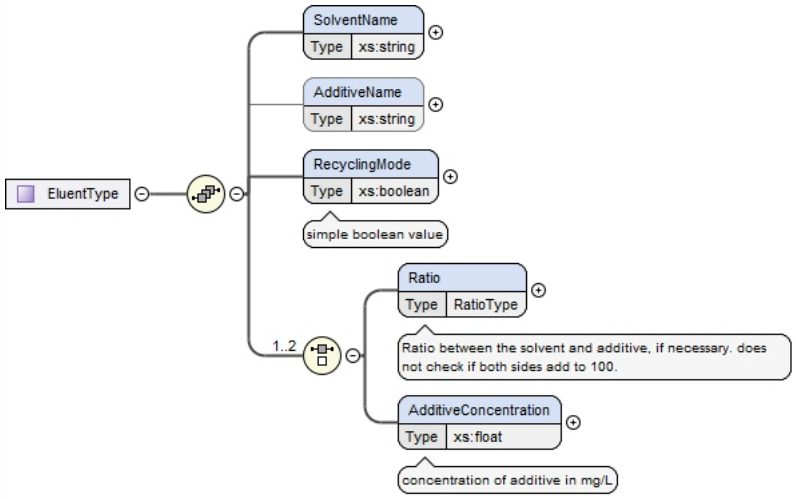
Namespace	No namespace			
Diagram				
Used by	Element	InstrumentConfigurationType/Pump		
Attributes	**QName**	**Type**	**Use**	
	FlowRate	PositiveFloatType	required	
		must be saved as mL/min		
Source	<xs:complexType name="PumpConfigurationType"> <xs:attribute name="FlowRate" type="PositiveFloatType" use="required"> <xs:annotation> <xs:documentation>must be saved as mL/min</xs:documentation> </xs:annotation> </xs:attribute> </xs:complexType>			

Complex Type `InjectorConfigurationType`

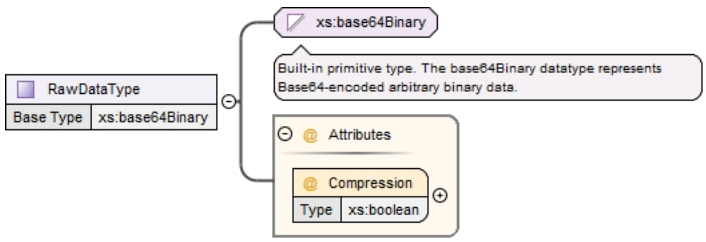
Namespace	No namespace			
Diagram				
Used by	Element	InstrumentConfigurationType/Injector		
Attributes	**QName**	**Type**	**Use**	
	InjectionVolume	PositiveFloatType	required	
		Must be saved in µL		
Source	<xs:complexType name="InjectorConfigurationType"> <xs:attribute name="InjectionVolume" type="PositiveFloatType" use="required"> <xs:annotation> <xs:documentation>Must be saved in µL</xs:documentation> </xs:annotation> </xs:attribute> </xs:complexType>			

Complex Type `EluentType`

Namespace	No namespace

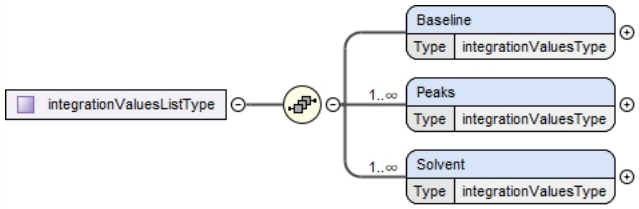
Diagram	
Used by	Element InstrumentConfigurationType/Eluent
Model	SolventName , AdditiveName{0,1} , RecyclingMode , (Ratio | AdditiveConcentration)
Children	AdditiveConcentration, AdditiveName, Ratio, RecyclingMode, SolventName
Source	<xs:complexType name="EluentType"> <xs:sequence> <xs:element name="SolventName" type="xs:string"/> <xs:element name="AdditiveName" type="xs:string" minOccurs="0"/> <xs:element name="RecyclingMode" type="xs:boolean"> <xs:annotation> <xs:documentation>simple boolean value</xs:documentation> </xs:annotation> </xs:element> <xs:choice minOccurs="1" maxOccurs="2"> <xs:element name="Ratio" type="RatioType"> <xs:annotation> <xs:documentation>Ratio between the solvent and additive, if necessary. does not check if both sides add to 100.</xs:documentation> </xs:annotation> </xs:element> <xs:element name="AdditiveConcentration" type="xs:float"> <xs:annotation> <xs:documentation>concentration of additive in mg/L</xs:documentation> </xs:annotation> </xs:element> </xs:choice> </xs:sequence> </xs:complexType>

Complex Type `RawDataType`

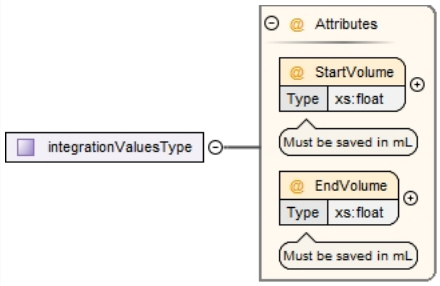
Namespace	No namespace
Diagram	
Type	extension of xs:base64Binary
Used by	Element RuntimeDataType/RawData
Attributes	**QName** | **Type** | **Use** **Compression** | xs:boolean | required
Source	<xs:complexType name="RawDataType"> <xs:simpleContent> <xs:extension base="xs:base64Binary"> <xs:attribute name="Compression" type="xs:boolean" use="required"/> </xs:extension>

```
  </xs:simpleContent>
</xs:complexType>
```

Complex Type `integrationValuesListType`

Namespace	No namespace
Diagram	
Used by	Element RuntimeDataType/integrationValues
Model	Baseline , Peaks+ , Solvent+
Children	Baseline, Peaks, Solvent
Source	<xs:complexType name="integrationValuesListType"> <xs:sequence> <xs:element name="Baseline" type="integrationValuesType"/> <xs:element name="Peaks" type="integrationValuesType" maxOccurs="unbounded"/> <xs:element name="Solvent" type="integrationValuesType" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType>

Complex Type `integrationValuesType`

Namespace	No namespace
Diagram	
Used by	Elements integrationValuesListType/Baseline, integrationValuesListType/Peaks, integrationValuesListType/Solvent

Attributes

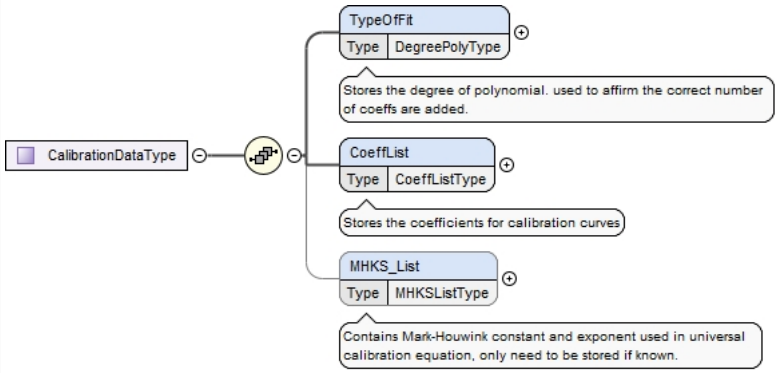
QName	Type	Use	
EndVolume	xs:float	required	
	Must be saved in mL		
StartVolume	xs:float	required	
	Must be saved in mL		

Source

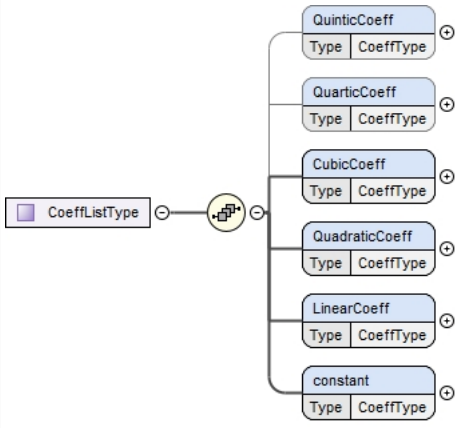
```
<xs:complexType name="integrationValuesType">
  <xs:attribute name="StartVolume" type="xs:float" use="required">
    <xs:annotation>
      <xs:documentation>Must be saved in mL</xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="EndVolume" type="xs:float" use="required">
    <xs:annotation>
      <xs:documentation>Must be saved in mL</xs:documentation>
    </xs:annotation>
  </xs:attribute>
</xs:complexType>
```

Complex Type `CalibrationDataType`

Namespace	No namespace

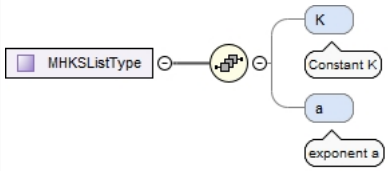
Diagram	
Used by	Element gpcMLType/CalibrationData
Model	TypeOfFit , CoeffList , MHKS_List{0,1}
Children	CoeffList, MHKS_List, TypeOfFit
Source	<xs:complexType name="CalibrationDataType"> <xs:sequence> <xs:element name="TypeOfFit" type="DegreePolyType"> <xs:annotation> <xs:documentation>Stores the degree of polynomial. used to affirm the correct number of coeffs are added.</xs:documentation> </xs:annotation> </xs:element> <xs:element name="CoeffList" type="CoeffListType"> <xs:annotation> <xs:documentation>Stores the coefficients for calibration curves</xs:documentation> </xs:annotation> </xs:element> <xs:element name="MHKS_List" type="MHKSListType" minOccurs="0"> <xs:annotation> <xs:documentation>Contains Mark-Houwink constant and exponent used in universal calibration equation, only need to be stored if known.</xs:documentation> </xs:annotation> </xs:element> </xs:sequence> </xs:complexType>

Complex Type `CoeffListType`

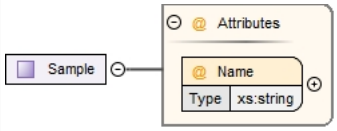
Namespace	No namespace
Diagram	
Used by	Element CalibrationDataType/CoeffList
Model	QuinticCoeff{0,1} , QuarticCoeff{0,1} , CubicCoeff , QuadraticCoeff , LinearCoeff , constant
Children	CubicCoeff, LinearCoeff, QuadraticCoeff, QuarticCoeff, QuinticCoeff, constant
Source	<xs:complexType name="CoeffListType"> <xs:sequence> <xs:element name="QuinticCoeff" type="CoeffType" minOccurs="0"/> <xs:element name="QuarticCoeff" type="CoeffType" minOccurs="0"/> <xs:element name="CubicCoeff" type="CoeffType"/> <xs:element name="QuadraticCoeff" type="CoeffType"/> <xs:element name="LinearCoeff" type="CoeffType"/> <xs:element name="constant" type="CoeffType"/>

```
  </xs:sequence>
</xs:complexType>
```

Complex Type MHKSListType

Namespace	No namespace
Annotations	
Diagram	
Used by	Element CalibrationDataType/MHKS_List
Model	K{0,1} , a{0,1}
Children	K, a
Source	<xs:complexType name="MHKSListType"> <xs:annotation> <xs:documentation/> </xs:annotation> <xs:sequence> <xs:element name="K" minOccurs="0"> <xs:annotation> <xs:documentation>Constant K</xs:documentation> </xs:annotation> </xs:element> <xs:element name="a" minOccurs="0"> <xs:annotation> <xs:documentation>exponent a</xs:documentation> </xs:annotation> </xs:element> </xs:sequence> </xs:complexType>

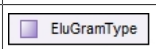
Complex Type Sample

Namespace	No namespace
Diagram	
Attributes	**QName** **Type** **Use** **Name** xs:string required
Source	<xs:complexType name="Sample"> <xs:attribute name="Name" type="xs:string" use="required"/> </xs:complexType>

Complex Type SampleType

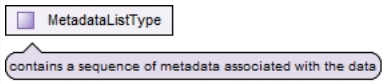
Namespace	No namespace
Diagram	
Source	<xs:complexType name="SampleType"/>

Complex Type EluGramType

Namespace	No namespace
Diagram	
Source	<xs:complexType name="EluGramType"/>

Complex Type MetadataListType

Namespace	No namespace

Annotations	contains a sequence of metadata associated with the data
Diagram	
Source	<xs:complexType name="MetadataListType"> <xs:annotation> <xs:documentation>contains a sequence of metadata associated with the data</xs:documentation> </xs:annotation> </xs:complexType>

Complex Type CalibrationType

Namespace	No namespace
Diagram	
Source	<xs:complexType name="CalibrationType"/>

Complex Type RIDetectorRunType

Namespace	No namespace
Diagram	
Source	<xs:complexType name="RIDetectorRunType"/>

Complex Type ViscometerRunType

Namespace	No namespace
Diagram	
Source	<xs:complexType name="ViscometerRunType"/>

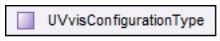
Complex Type UVvisRunType

Namespace	No namespace
Diagram	
Source	<xs:complexType name="UVvisRunType"/>

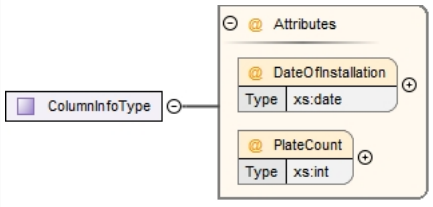
Complex Type LightScatteringRunType

Namespace	No namespace
Diagram	
Source	<xs:complexType name="LightScatteringRunType"/>

Complex Type UVvisConfigurationType

Namespace	No namespace
Diagram	
Source	<xs:complexType name="UVvisConfigurationType"/>

Complex Type ColumnInfoType

Namespace	No namespace
Diagram	

Attributes	QName	Type	Use
	DateOfInstallation	xs:date	optional
	PlateCount	xs:int	optional

Source

```
<xs:complexType name="ColumnInfoType">
  <xs:attribute name="DateOfInstallation" type="xs:date" use="optional"/>
  <xs:attribute name="PlateCount" type="xs:int" use="optional"/>
</xs:complexType>
```

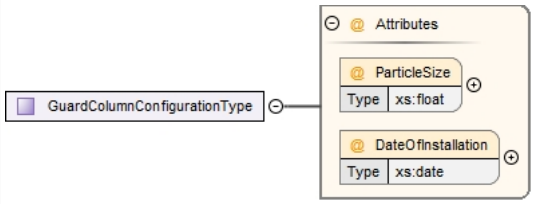
Complex Type `RIDetectorConfigurationType`

Namespace	No namespace
Diagram	
Source	`<xs:complexType name="RIDetectorConfigurationType"/>`

Complex Type `LightScatteringConfigurationType`

Namespace	No namespace
Diagram	
Source	`<xs:complexType name="LightScatteringConfigurationType"/>`

Complex Type `GuardColumnConfigurationType`

Namespace	No namespace
Diagram	

Attributes	QName	Type	Use
	DateOfInstallation	xs:date	optional
	ParticleSize	xs:float	optional

Source

```
<xs:complexType name="GuardColumnConfigurationType">
  <xs:attribute name="ParticleSize" type="xs:float" use="optional"/>
  <xs:attribute name="DateOfInstallation" type="xs:date"/>
</xs:complexType>
```

Complex Type `PolyCoeffType`

Namespace	No namespace
Diagram	
Source	`<xs:complexType name="PolyCoeffType"/>`

Complex Type `StandardUsedType`

Namespace	No namespace
Diagram	
Source	`<xs:complexType name="StandardUsedType"/>`

Complex Type `RangeOfStandardsType`

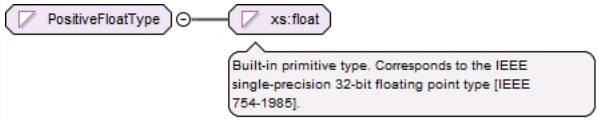
Namespace	No namespace
Diagram	
Source	`<xs:complexType name="RangeOfStandardsType"/>`

Complex Type `CalibrationDateType`

Namespace	No namespace
Diagram	
Source	`<xs:complexType name="CalibrationDateType"/>`

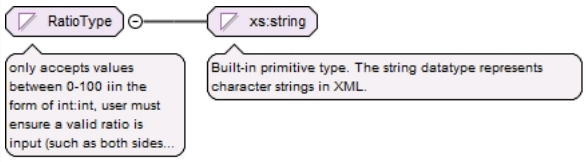
Simple Type(s)

Simple Type `PositiveFloatType`

Namespace	No namespace
Diagram	
Type	restriction of xs:float
Facets	minInclusive `0`
Used by	Attributes InjectorConfigurationType/@InjectionVolume, OvenConfigurationType/@Temperature, PumpConfigurationType/@FlowRate

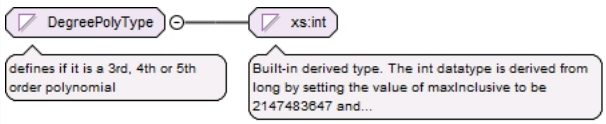
Source	`<xs:simpleType name="PositiveFloatType">`
	`  <xs:restriction base="xs:float">`
	`  <xs:minInclusive value="0"/>`
	`  </xs:restriction>`
	`</xs:simpleType>`

Simple Type `RatioType`

Namespace	No namespace
Annotations	only accepts values between 0-100 iin the form of int:int, user must ensure a valid ratio is input (such as both sides adding to 100)
Diagram	
Type	restriction of xs:string
Facets	pattern `(100|[0-9]?[0-9]):(100|[0-9]?[0-9])`
Used by	Element EluentType/Ratio

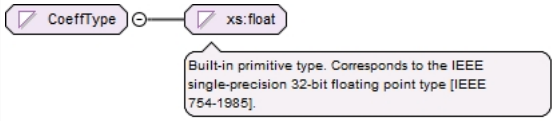
Source	`<xs:simpleType name="RatioType">`
	`  <xs:annotation>`
	`  <xs:documentation>only accepts values between 0-100 iin the form of int:int, user must ensure a valid ratio is input (such as both sides adding to 100)</xs:documentation>`
	`  </xs:annotation>`
	`  <xs:restriction base="xs:string">`
	`  <xs:pattern value="(100|[0-9]?[0-9]):(100|[0-9]?[0-9])"/>`
	`  </xs:restriction>`
	`</xs:simpleType>`

Simple Type `DegreePolyType`

Namespace	No namespace
Annotations	defines if it is a 3rd, 4th or 5th order polynomial
Diagram	
Type	restriction of xs:int

Facets	maxInclusive	5
	minInclusive	3
Used by	Element	CalibrationDataType/TypeOfFit
Source	<xs:simpleType name="DegreePolyType"> <xs:annotation> <xs:documentation>defines if it is a 3rd, 4th or 5th order polynomial</xs:documentation> </xs:annotation> <xs:restriction base="xs:int"> <xs:minInclusive value="3"/> <xs:maxInclusive value="5"/> </xs:restriction> </xs:simpleType>	

Simple Type CoeffType

Namespace	No namespace
Diagram	
Type	xs:float
Used by	Elements: CoeffListType/CubicCoeff, CoeffListType/LinearCoeff, CoeffListType/QuadraticCoeff, CoeffListType/QuarticCoeff, CoeffListType/QuinticCoeff, CoeffListType/constant
Source	<xs:simpleType name="CoeffType"> <xs:restriction base="xs:float"/> </xs:simpleType>

Attribute(s)

Attribute ContactType / @FirstName

Namespace	No namespace	
Type	xs:string	
Properties	use:	required
Used by	Complex Type	ContactType
Source	<xs:attribute name="FirstName" type="xs:string" use="required"/>	

Attribute ContactType / @MiddleName

Namespace	No namespace	
Type	xs:string	
Properties	use:	optional
Used by	Complex Type	ContactType
Source	<xs:attribute name="MiddleName" type="xs:string" use="optional"/>	

Attribute ContactType / @LastName

Namespace	No namespace	
Type	xs:string	
Properties	use:	required
Used by	Complex Type	ContactType
Source	<xs:attribute name="LastName" type="xs:string" use="required"/>	

Attribute ContactType / @E-mail

Namespace	No namespace
Type	xs:string

Properties	content:	simple
Used by	Complex Type	ContactType
Source	<xs:attribute name="E-mail" type="xs:string"/>	

Attribute ContactType / @Institution

Namespace	No namespace	
Type	xs:string	
Properties	use:	optional
Used by	Complex Type	ContactType
Source	<xs:attribute name="Institution" type="xs:string" use="optional"/>	

Attribute MetadataType / @date

Namespace	No namespace	
Type	xs:date	
Properties	use:	required
Used by	Complex Type	MetadataType
Source	<xs:attribute name="date" type="xs:date" use="required"/>	

Attribute MetadataType / @MethodName

Namespace	No namespace	
Type	xs:string	
Properties	use:	required
Used by	Complex Type	MetadataType
Source	<xs:attribute name="MethodName" type="xs:string" use="required"/>	

Attribute MetadataType / @CalibrationName

Namespace	No namespace	
Type	xs:string	
Properties	use:	optional
Used by	Complex Type	MetadataType
Source	<xs:attribute name="CalibrationName" type="xs:string" use="optional"/>	

Attribute MetadataType / @Operator

Namespace	No namespace
Annotations	If multiple contacts, state name (first-last) of who ran the sample
Type	xs:string
Properties	content: simple
Used by	Complex Type MetadataType
Source	<xs:attribute name="Operator" type="xs:string"> <xs:annotation> <xs:documentation>If multiple contacts, state name (first-last) of who ran the sample</ xs:documentation> </xs:annotation> </xs:attribute>

Attribute MetadataType / @InstrumentName

Namespace	No namespace

Type	xs:string
Properties	use: required
Used by	Complex Type MetadataType
Source	`<xs:attribute name="InstrumentName" type="xs:string" use="required"/>`

Attribute MetadataType / @RunTime

Namespace	No namespace
Annotations	Length of the run sample in Minutes
Type	xs:time
Properties	use: optional
Used by	Complex Type MetadataType

Source	`<xs:attribute name="RunTime" type="xs:time" use="optional">`
	`  <xs:annotation>`
	`  <xs:documentation>Length of the run sample in Minutes</xs:documentation>`
	`  </xs:annotation>`
	`</xs:attribute>`

Attribute MetadataType / @lastServiceDate

Namespace	No namespace
Type	xs:date
Properties	use: optional
Used by	Complex Type MetadataType
Source	`<xs:attribute name="lastServiceDate" type="xs:date" use="optional"/>`

Attribute MetadataType / @SampleName

Namespace	No namespace
Type	xs:string
Properties	use: required
Used by	Complex Type MetadataType
Source	`<xs:attribute name="SampleName" type="xs:string" use="required"/>`

Attribute MetadataType / @ObservationalData

Namespace	No namespace
Annotations	Add comments relating to topology, functionality, monomer composition, polymer type, ect
Type	xs:string
Properties	content: simple
Used by	Complex Type MetadataType

Source	`<xs:attribute name="ObservationalData" type="xs:string">`
	`  <xs:annotation>`
	`  <xs:documentation>Add comments relating to topology, functionality, monomer composition, polymer type, ect</xs:documentation>`
	`  </xs:annotation>`
	`</xs:attribute>`

Attribute InstrumentInfoType / @Name

Namespace	No namespace
Type	xs:string
Properties	use: optional
Used by	Complex Type InstrumentInfoType

Source	`<xs:attribute name="Name" type="xs:string" use="optional"/>`

Attribute `InstrumentInfoType / @SerialNumber`

Namespace	No namespace
Annotations	for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.
Type	xs:string
Properties	content: simple
Used by	Complex Type InstrumentInfoType
Source	`<xs:attribute name="SerialNumber" type="xs:string"><xs:annotation><xs:documentation>for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.</xs:documentation></xs:annotation></xs:attribute>`

Attribute `InstrumentInfoType / @Manufacturer`

Namespace	No namespace
Annotations	for saving what columns were used
Type	xs:string
Properties	use: required
Used by	Complex Type InstrumentInfoType
Source	`<xs:attribute name="Manufacturer" type="xs:string" use="required"><xs:annotation><xs:documentation>for saving what columns were used</xs:documentation></xs:annotation></xs:attribute>`

Attribute `ColumnConfigurationType / @PoreSizeMin`

Namespace	No namespace
Type	xs:positiveInteger
Properties	use: required
Used by	Complex Type ColumnConfigurationType
Source	`<xs:attribute name="PoreSizeMin" type="xs:positiveInteger" use="required"/>`

Attribute `ColumnConfigurationType / @PoreSizeMax`

Namespace	No namespace
Type	xs:positiveInteger
Properties	use: required
Used by	Complex Type ColumnConfigurationType
Source	`<xs:attribute name="PoreSizeMax" type="xs:positiveInteger" use="required"/>`

Attribute `ColumnConfigurationType / @StationaryPhase`

Namespace	No namespace
Type	xs:string
Properties	use: optional
Used by	Complex Type ColumnConfigurationType
Source	`<xs:attribute name="StationaryPhase" type="xs:string" use="optional"/>`

Attribute `ColumnConfigurationType / @DateOfInstallation`

Namespace	No namespace

Type	xs:date
Properties	use: optional
Used by	Complex Type ColumnConfigurationType
Source	`<xs:attribute name="DateOfInstallation" type="xs:date" use="optional"/>`

Attribute `OvenConfigurationType / @Temperature`

Namespace	No namespace
Annotations	Must be saved in celcius
Type	PositiveFloatType
Properties	use: required
Facets	minInclusive 0
Used by	Complex Type OvenConfigurationType

Source	`<xs:attribute name="Temperature" type="PositiveFloatType" use="required">`
	`  <xs:annotation>`
	`  <xs:documentation>Must be saved in celcius</xs:documentation>`
	`  </xs:annotation>`
	`</xs:attribute>`

Attribute `PumpConfigurationType / @FlowRate`

Namespace	No namespace
Annotations	must be saved as mL/min
Type	PositiveFloatType
Properties	use: required
Facets	minInclusive 0
Used by	Complex Type PumpConfigurationType

Source	`<xs:attribute name="FlowRate" type="PositiveFloatType" use="required">`
	`  <xs:annotation>`
	`  <xs:documentation>must be saved as mL/min</xs:documentation>`
	`  </xs:annotation>`
	`</xs:attribute>`

Attribute `InjectorConfigurationType / @InjectionVolume`

Namespace	No namespace
Annotations	Must be saved in µL
Type	PositiveFloatType
Properties	use: required
Facets	minInclusive 0
Used by	Complex Type InjectorConfigurationType

Source	`<xs:attribute name="InjectionVolume" type="PositiveFloatType" use="required">`
	`  <xs:annotation>`
	`  <xs:documentation>Must be saved in µL</xs:documentation>`
	`  </xs:annotation>`
	`</xs:attribute>`

Attribute `RawDataType / @Compression`

Namespace	No namespace
Type	xs:boolean
Properties	use: required
Used by	Complex Type RawDataType
Source	`<xs:attribute name="Compression" type="xs:boolean" use="required"/>`

Attribute integrationValuesType / @StartVolume

Namespace	No namespace
Annotations	Must be saved in mL
Type	xs:float
Properties	use: required
Used by	Complex Type integrationValuesType
Source	<xs:attribute name="StartVolume" type="xs:float" use="required"> <xs:annotation> <xs:documentation>Must be saved in mL</xs:documentation> </xs:annotation> </xs:attribute>

Attribute integrationValuesType / @EndVolume

Namespace	No namespace
Annotations	Must be saved in mL
Type	xs:float
Properties	use: required
Used by	Complex Type integrationValuesType
Source	<xs:attribute name="EndVolume" type="xs:float" use="required"> <xs:annotation> <xs:documentation>Must be saved in mL</xs:documentation> </xs:annotation> </xs:attribute>

Attribute gpcMLType / @Version

Namespace	No namespace
Annotations	gpcML version used.
Type	xs:string
Properties	use: required
Used by	Complex Type gpcMLType
Source	<xs:attribute name="Version" type="xs:string" use="required"> <xs:annotation> <xs:documentation>gpcML version used.</xs:documentation> </xs:annotation> </xs:attribute>

Attribute gpcMLType / @id

Namespace	No namespace
Annotations	id associated with the gpcML document
Type	xs:int
Properties	content: simple
Used by	Complex Type gpcMLType
Source	<xs:attribute name="id" type="xs:int"> <xs:annotation> <xs:documentation>id associated with the gpcML document</xs:documentation> </xs:annotation> </xs:attribute>

Attribute Sample / @Name

Namespace	No namespace
Type	xs:string
Properties	use: required
Used by	Complex Type Sample

Source	`<xs:attribute name="Name" type="xs:string" use="required"/>`

Attribute `ColumnInfoType / @DateOfInstallation`

Namespace	No namespace	
Type	xs:date	
Properties	use:	`optional`
Used by	Complex Type	ColumnInfoType
Source	`<xs:attribute name="DateOfInstallation" type="xs:date" use="optional"/>`	

Attribute `ColumnInfoType / @PlateCount`

Namespace	No namespace	
Type	xs:int	
Properties	use:	`optional`
Used by	Complex Type	ColumnInfoType
Source	`<xs:attribute name="PlateCount" type="xs:int" use="optional"/>`	

Attribute `GuardColumnConfigurationType / @ParticleSize`

Namespace	No namespace	
Type	xs:float	
Properties	use:	`optional`
Used by	Complex Type	GuardColumnConfigurationType
Source	`<xs:attribute name="ParticleSize" type="xs:float" use="optional"/>`	

Attribute `GuardColumnConfigurationType / @DateOfInstallation`

Namespace	No namespace	
Type	xs:date	
Properties	content:	`simple`
Used by	Complex Type	GuardColumnConfigurationType
Source	`<xs:attribute name="DateOfInstallation" type="xs:date"/>`	